Aspiration-Weighted Influence

Siming Ye*

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Abstract

When consumers look to others' purchases for guidance, observed choices can reflect both personal preferences and external influence. This paper asks how to disentangle these components when some observed purchases are unavailable to the consumer. I develop a model that combines the consumer's idiosyncratic choice tendencies, which describe how she would choose without influence from this source, with the influencer's relative purchase frequencies among available goods. The weight on the influencer's frequencies depends on a stable influence parameter and on the share of observed purchases the consumer can follow. A behavioral characterization makes the decomposition testable, and I establish when variation in the influencer's purchases identifies both the consumer's idiosyncratic choice tendencies and her influence parameter. The model predicts that the same change in relative purchase frequencies among available goods has a larger effect on consumer choice when these goods account for a larger share of the influencer's purchases.

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1 Introduction

Consumers often look to other people's purchases for guidance, raising the question of what their observed choices reveal about their own preferences and their response to others.[1] Before buying an item of clothing, for example, a consumer might look at the purchasing pattern of an influencer, such as a fashion blogger, to see what is trendy. Location, budget, or time constraints may mean that she chooses from a set of goods that differs from the blogger's. Her idiosyncratic choice tendencies describe how she would choose without this source of influence. This paper asks when the consumer's choices, together with the influencer's observed purchases, allow these tendencies to be disentangled from her response to a purchasing pattern she can only partly follow.

Disentangling these components clarifies what observed purchases reveal about individual preferences and how influence changes consumption. Frequent selection of a good favored by the influencer can reflect a strong idiosyncratic tendency combined with a modest response, or a weaker tendency reinforced by a stronger response. These accounts imply different choices without the source and different adjustments when the influencer begins favoring other goods.

The challenge of disentangling individual preferences from external influence has been examined from several perspectives in economics. Manski (1993) studies how to distinguish responses to group behavior from effects of group characteristics and shared environments when individual and group behavior are jointly determined. Brock and Durlauf (2001) study equilibrium and identification in a discrete-choice model with private and social components of utility. Chambers et al. (2023) provide a behavioral characterization and identification results for a stochastic-choice model of peer influence. These approaches use different restrictions and sources of variation to separate individual and social contributions to observed choice. In the present setting, a single purchasing pattern leaves multiple combinations of idiosyncratic choice tendencies and responsiveness consistent with the consumer's choices.

Separating these explanations requires restrictions that link behavior across different purchasing patterns. This paper takes a decision-theoretic approach, placing those restrictions on the consumer's observable responses while her available goods remain fixed. Each proposed decomposition must explain how she responds when the influencer shifts purchases among available goods or between available and unavailable goods. The same idiosyncratic choice tendencies and responsiveness must account for choices across these patterns.

[1] Huh et al. (2014) show experimentally that observing another consumer's choice can make the same option more likely to be chosen.

The consumer's behavior is described by the probability of choosing each available good, while the influencer's purchasing pattern is recorded by relative purchase frequencies over all observed goods. Within this stochastic choice framework, aspiration-weighted influence (AWI) combines idiosyncratic choice tendencies with the influencer's relative purchase frequencies among available goods. The weight on the influencer's frequencies depends on a stable influence parameter and the share of observed purchases involving available goods, called the available share.

The characterization gives necessary and sufficient observable conditions for AWI to describe choices across purchasing patterns. Within this model, two patterns yielding different consumer choice probabilities uniquely determine the consumer's idiosyncratic choice tendencies and influence parameter when these components and her available goods remain stable. Her choices under influence therefore reveal how she would choose without the source. Choices under additional purchasing patterns test whether the recovered components continue to account for her behavior.

The decomposition shows how stable responsiveness and changes in the available share jointly shape consumption. Suppose a good would be chosen 80 percent of the time without this source, while it accounts for 50 percent of the influencer's purchases among available goods. With the consumer's available goods, parameters, and the influencer's relative purchase frequencies among available goods held fixed, a higher available share moves her choice probability toward 50 percent. The good accounts for a larger share of the influencer's purchases while the consumer chooses it less often. The same structure also predicts that a given change in relative purchase frequencies among available goods elicits a larger response at a higher available share.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 sets out the choice environment and introduces AWI. Section 4 examines how the available share shapes the consumer's response to observed purchases. Section 5 characterizes the model on a fixed menu and develops the extension across menus. Section 6 establishes when variation in the influencer's observed purchases identifies the consumer's idiosyncratic choice tendencies and influence parameter. Section 7 analyzes the influencer's optimal choice of which purchases to display. Section 8 concludes. The appendices contain proofs and additional results.

2 Related Literature

A stochastic choice rule records how often each alternative is selected from every menu of available alternatives. Luce's choice axiom and McFadden's random-utility formulation showed how

restrictions on these probabilities can reveal an underlying utility structure (Luce, 1959; McFadden, 1974). Subsequent work has given randomness several different foundations. It may reflect a distribution over preference orderings, a perturbation of utility, a limited consideration set, or costly information acquisition (Gul and Pesendorfer, 2006; Fudenberg et al., 2015; Masatlioglu et al., 2012; Manzini and Mariotti, 2014; Matějka and McKay, 2015). Chambers et al. (2025) assign each alternative a utility and a salience parameter reflecting its prominence to the decision maker. Azrieli and Rehbeck (2026) characterize what marginal distributions of menus and of choices reveal when conditional choice frequencies go unobserved.

A second strand treats an observed choice rule as a mixture of component rules and asks what the mixture reveals. Turansick (2026) characterizes which distributions over a restricted set of preference types are consistent with aggregate choice frequencies. Suleymanov (2026) studies an agent whose choices blend a principal's Luce rule with its own and shows when the degree of alignment between the two is identified from the agent's choices alone. Chambers et al. (2024) study when joint choice can be explained by a common latent variable, conditional on which each agent's choice depends only on her own menu.

Most of this literature takes the menu as the principal source of variation. An influencer can change which goods she purchases or displays while the consumer continues to choose from the same menu. These changes provide variation in the exposure distribution with the consumer's menu held fixed.

Choice-theoretic models of social influence study how another person's action, a social norm, or a distribution of peer behavior affects individual choice (Cuhadaroglu, 2017; Chambers et al., 2023; Borah and Kops, 2019; Volpe, 2025). Social observations may alter preferences, consideration, or both (Kashaev and Lazzati, 2021; Kashaev et al., 2026). These models commonly give the decision maker and the social source a shared opportunity set, an assumption that fits settings in which peers choose from the same alternatives. An influencer can reach consumers whose budgets, locations, and eligibility give them different menus from the influencer's.

Peer behavior is often correlated with common shocks, sorting, and endogenous group formation. Those concerns motivate network instruments, random assignment, and experimentally induced information (Manski, 1993; Sacerdote, 2001; Bursztyn et al., 2014; Blume et al., 2011, 2015; Bramoullé et al., 2020; Boucher et al., 2024). Here, the consumer observes the influencer's choices before making her own. The identification analysis holds her menu, baseline distribution, and influence parameter fixed while these observations vary. Frequent selection of an alternative can reflect

either a high baseline probability or a response to its frequency in the influencer's observed choices. Variation in those observations separates the two under the maintained stability conditions.

Revealed-attention and random-consideration models allow an available alternative to go unchosen because the consumer does not consider it (Masatlioglu et al., 2012; Manzini and Mariotti, 2014). A consumer can also observe an alternative outside her menu. The photographer, for example, may watch a review of a lens unavailable to her and remember it vividly.

Phantom alternatives alter evaluation after they disappear from the feasible set (Farquhar and Pratkanis, 1993). In aspiration-based choice, an unavailable aspiration guides the decision maker toward a nearby feasible alternative, and restriction-sensitive choice lets a forbidden option redirect demand within the available set (Güney and Richter, 2015; Güney et al., 2018; Boissonnet and Ghersengorin, 2026). In these models, the identity or characteristics of an unavailable alternative affect choices among available alternatives.

Social platforms combine product discovery, information, advertising, and affiliation, while consumers differ in their budgets and access to goods (Aridor et al., 2024). An influencer choosing what to display therefore faces an audience whose members can choose different goods. The return from a display depends on which consumers observe it and which goods are available to them.

3 The Model

3.1 Primitives

Let X be a finite set of alternatives and let $S \subseteq X$ be the nonempty menu from which the consumer can choose. Alternatives in S are *available* to the consumer; those outside S are *unavailable* to her. In the purchase interpretation, alternatives are goods; the menu may also include an outside option representing no purchase. Write $\Delta(X)$ and $\Delta(S)$ for the probability distributions on X and S, respectively.

Availability describes what the consumer can choose on the current occasion. When the only restriction is her budget, the available goods are those priced within that budget. Stockouts, location, or eligibility restrictions can also make a good unavailable even when its price is within her budget. These conditions, and hence the menu, remain fixed as the influencer's observed choices vary.

The consumer observes choices made by an external agent, called the *influencer*. The *exposure distribution*, $q \in \Delta(X)$, records the relative frequencies of these observed choices. For each $x \in X$,

$q(x)$ is the fraction of observations in which the influencer chooses x. In the content application, each observation is a display of a product the influencer consumes. The influencer has access to each alternative she is observed choosing; some of these alternatives may lie outside the consumer's menu S. Her observed choices are the source of variation in q.

The vector $q|_S = (q(x))_{x \in S}$ records the observed frequencies of alternatives in S. Each frequency is measured among all observed influencer choices, without renormalizing over S. Its total mass, $q_S \equiv \sum_{x \in S} q(x)$, is the *available share*: the fraction of the influencer's observed choices that the consumer can also make. When $q_S > 0$, the *conditional exposure distribution* is $q(x \mid S) = q(x)/q_S$ for $x \in S$. It gives the relative frequencies within those available choices. The remaining share, $1 - q_S$, consists of choices outside the consumer's menu. If the influencer's choices all belong to S, then $q_S = 1$.

In the absence of influence from this agent, the consumer chooses each $x \in S$ with probability $p_0(x \mid S)$. These probabilities describe her idiosyncratic choice tendencies and form her *baseline distribution*, $p_0(\cdot \mid S) \in \Delta(S)$. It remains fixed as q varies on a given menu. Let $p(\cdot \mid S; q) \in \Delta(S)$ denote her choice probabilities when she observes the influencer's choices with frequencies q. The analysis concerns one consumer; the same probabilities also describe a population whose members share this menu, baseline, and response to the same influencer.

3.2 The Choice Rule

A repeated sampling procedure gives one interpretation of the choice rule. Let $\alpha \in (0, 1)$ be the consumer's *influence parameter*. On each attempt, with probability $1 - \alpha$ she selects according to her baseline: each $x \in S$ is selected with probability $p_0(x \mid S)$. With probability α, she imitates a randomly selected observation of the influencer: each $x \in \mathcal{X}$ is selected with probability $q(x)$. If the selected alternative belongs to S, the consumer chooses it and the procedure ends. Otherwise, she makes another independent attempt with the same baseline, exposure distribution, and influence parameter.

Extend $p_0(\cdot \mid S)$ by zero outside S, and let $M(\cdot \mid S; q)$ denote the distribution of the alternative drawn on an attempt:

$$M(\cdot \mid S; q) = (1 - \alpha) p_0(\cdot \mid S) + \alpha q.$$

The probability that an attempt completes the choice is $(1 - \alpha) + \alpha q_S$. It is strictly positive because $\alpha < 1$, so a choice is eventually completed with probability one. The choice distribution is M

conditional on S. For each $x \in S$,

$$p(x \mid S; q) = \frac{(1-\alpha)p_0(x \mid S) + \alpha q(x)}{(1-\alpha) + \alpha q_S}, \qquad x \in S. \tag{1}$$

Equation (1) defines the *aspiration-weighted influence* (AWI) choice rule. The characterization and identification results concern this distribution of completed choices. Several procedures can generate the same distribution, so an interpretation of the intermediate steps requires additional assumptions. If $q_S = 0$, completed choices follow the baseline. If $q_S = 1$, the rule is the mixture $(1-\alpha)p_0 + \alpha q$.

Under the repeated sampling procedure, the denominator in (1) is the completion probability on each attempt. The expected number of attempts is therefore $[(1-\alpha) + \alpha q_S]^{-1}$. A lower available share increases this expected completion time and reduces the fraction of completed choices that result from imitating the influencer. Section 7 adopts this procedure when assigning a cost to delayed completion.

Figure 1 shows this sampling interpretation. The distribution M combines the baseline and exposure distributions over $\mathcal{X}$. Conditioning M on S gives the distribution of completed choices.

For $q_S > 0$, conditioning the exposure distribution on S before mixing would give $(1-\alpha)p_0(\cdot \mid S) + \alpha q(\cdot \mid S)$. In that rule, the conditional exposure distribution has weight α at every available share. It agrees with (1) exactly when $q_S = 1$ or $q(\cdot \mid S) = p_0(\cdot \mid S)$. Appendix B compares the choice distributions generated by alternative sampling procedures.

Across menus, let $u : \mathcal{X} \to \mathbb{R}_{++}$ be a positive Luce index, where $u(x)$ represents the consumer's personal value for alternative x. A *Luce baseline distribution* assigns each $x \in S$ probability proportional to $u(x)$, normalized by the sum over alternatives $y \in S$:

$$p_0(x \mid S) = \frac{u(x)}{\sum_{y \in S} u(y)}, \qquad x \in S. \tag{2}$$

Combined with the same influence parameter α on every menu, this baseline gives the *Aspiration-Weighted Luce Model*. Section 5.2 provides conditions for this cross-menu specification.

4 Comparative Statics

The influencer's observed purchases have two relevant features: how often they involve available goods, and how they are distributed among those goods. Write $\kappa = (1-\alpha)/\alpha$. Dividing the

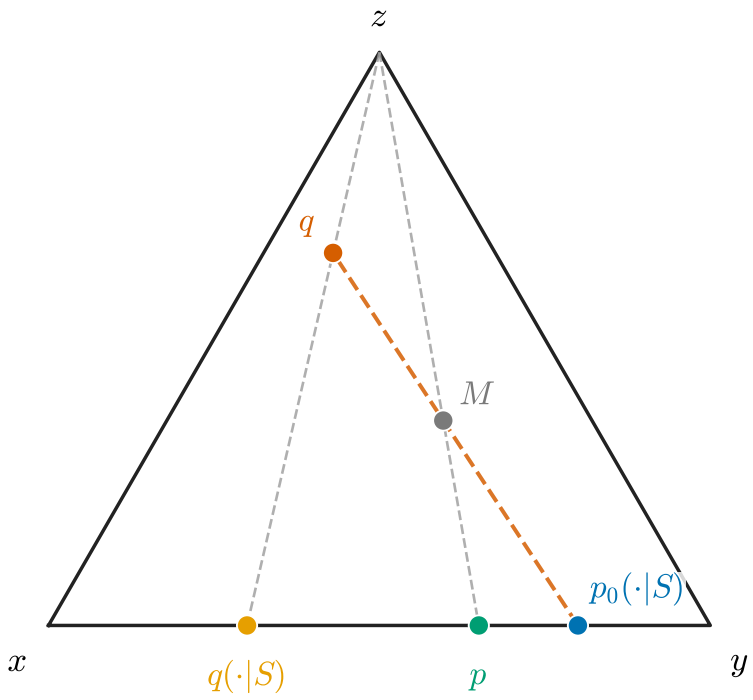


Figure 1: Aspiration-weighted influence. The attempt distribution M combines the baseline and exposure distributions. Conditioning M on the consumer's menu produces the observed choice distribution p.

numerator and denominator of (1) by α gives

$$p(\cdot \mid S; q) = \frac{\kappa p_0(\cdot \mid S) + q|_S}{\kappa + q_S}. \tag{3}$$

For an available share $\mu \in [0, 1]$, write $\beta(\mu)$ for the weight on the conditional exposure distribution. When $q_S > 0$, substituting $q|_S = q_S q(\cdot \mid S)$ gives

$$p(\cdot \mid S; q) = \big[1 - \beta(q_S)\big] p_0(\cdot \mid S) + \beta(q_S) q(\cdot \mid S), \qquad \beta(\mu) = \frac{\alpha\mu}{(1-\alpha) + \alpha\mu}. \tag{4}$$

The weight $\beta(q_S)$ increases with the available share and equals α when all observed purchases are available. A smaller available share can therefore reduce responsiveness even with an unchanged baseline and influence parameter. Under the sampling interpretation, $\beta(q_S)$ is the probability that the completed choice resulted from imitation. Whenever the influencer is observed choosing goods outside the consumer's menu, $\beta(q_S) < \alpha$, because an imitation attempt can select an unavailable good.

Fix the menu, baseline distribution, and influence parameter. Suppose the influencer's observed purchases change from exposure distribution q to q', with the same positive available share μ. The consumer's choice probabilities then change by

$$p(\cdot \mid S; q') - p(\cdot \mid S; q) = \beta(\mu)\big[q'(\cdot \mid S) - q(\cdot \mid S)\big].$$

The same change in relative frequencies among available goods therefore produces a larger response at a higher available share. For the photographer, suppose the influencer changes the purchase split between two available lenses from 50:50 to 70:30. Holding the menu, baseline, and influence parameter fixed, this change has a larger effect when available lenses account for 80 percent of observed purchases than when they account for 20 percent.

An influencer who displays available goods more often can therefore induce a larger demand response to the same change in their relative frequencies, holding the consumer's menu, baseline distribution, and influence parameter fixed.

Redistributing observed purchases among unavailable goods leaves consumer choice unchanged. Their total share affects how far choice moves from the baseline. Holding the relative frequencies of available purchases fixed, a lower available share moves choice toward the baseline.

Proposition 4.1 (Aspirational dampening)**.** *Fix S, $p_0(\cdot \mid S)$, and $\alpha \in (0, 1)$. Let q and q' have the same conditional exposure distribution on S, with positive available shares. Then*

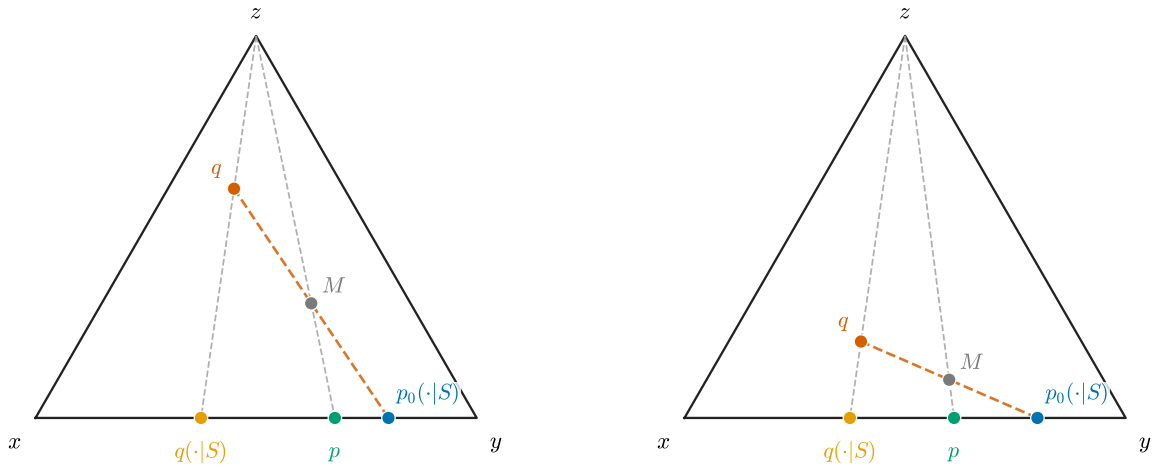
$$p(\cdot \mid S; q') - p(\cdot \mid S; q) = \big[\beta(q'_S) - \beta(q_S)\big]\big[q(\cdot \mid S) - p_0(\cdot \mid S)\big]. \tag{5}$$

If $q'_S < q_S$, choice under q' lies between $p_0(\cdot \mid S)$ and choice under q on the segment joining the baseline to their common conditional exposure distribution.

A fall in the available share leaves the direction of influence intact and reduces its magnitude. Coordinate by coordinate,

$$p(x \mid S; q) - p_0(x \mid S) = \beta(q_S)\big[q(x \mid S) - p_0(x \mid S)\big].$$

Every gap from baseline contracts by the same factor. An alternative becomes less likely to be chosen if its probability in the conditional exposure distribution exceeds its baseline probability, and more likely if that probability is lower. Figure 2 shows the resulting movement toward the consumer's baseline as the available share falls.



(a) Available share $q_S = 0.4$

(b) Available share $q_S = 0.8$

Figure 2: The conditional exposure distribution is the same in both panels. A larger available share increases its weight and moves the choice distribution farther from the baseline distribution.

The one-shot fallback and finite-cap rules in Appendix B also move choice toward the baseline when the available share falls. These rules differ in how the weight on the conditional exposure distribution varies with that share.

5 Characterization

Three restrictions on responses to changes in exposure characterize aspiration-weighted influence on a fixed menu. Two additional comparisons across menus yield a common influence parameter and a positive Luce index.

5.1 A fixed menu

Fix a proper menu $S \subsetneq X$ with $|S| \geq 3$, where X is the finite universe of alternatives. Retain $p(x \mid S; q)$ for the probability of choosing x and $q_S = \sum_{x \in S} q(x)$ for the available share. The observed rule is defined on $Q_S = \{q \in \Delta(X) : q_S > 0\}$, with $p(\cdot \mid S; q) \in \Delta(S)$. The domain includes zero exposure coordinates and exposure distributions supported entirely on S. Let δ_x denote exposure concentrated on x. For distinct $a, b \in X$, the exposure distribution $q + t(\delta_b - \delta_a)$ transfers t units of exposure from a to b, leaving every other coordinate fixed. It is feasible for $0 < t \leq q(a)$. The first two axioms apply to every exposure distribution and every feasible transfer in their stated domains.

We compare two kinds of transfers. The first reallocates exposure between available alternatives, keeping all unavailable exposure fixed. The second replaces unavailable exposure with exposure to an available alternative. Both comparisons keep the menu fixed and concern choices among alternatives whose own exposure has not changed. To illustrate, consider a photographer choosing among three available lenses x, y, k, while the influencer's observed purchases also include an unavailable lens z.

Axiom 1 (Exposure Reallocation Independence)**.** For every $q \in Q_S$, $x \in S$, distinct $a, b \in S \setminus \{x\}$, and $0 < t \leq q(a)$,

$$p\big(x \mid S; q + t(\delta_b - \delta_a)\big) = p(x \mid S; q). \tag{6}$$

Holding an alternative's own exposure and the entire unavailable-exposure vector fixed, its choice probability is independent of how exposure is distributed among the other available alternatives. Any such reallocation can be made through finitely many of the stated transfers. Since choice probabilities sum to one, the probability changes of the two affected alternatives are equal and opposite. For the photographer, replacing some observed purchases of lens y with purchases of lens k leaves the probability of choosing x unchanged, with observed purchase frequencies of x and every unavailable lens held fixed.

For distinct $x, y \in S$, *relative choice* is the fraction of choices of x among choices of x or y from the original menu S. It equals $p(x \mid S; q)/[p(x \mid S; q) + p(y \mid S; q)]$ when the denominator is positive.

Axiom 2 (Relative Choice Independence)**.** For every $q \in Q_S$, distinct $x, y \in S$, $k \in S \setminus \{x, y\}$, $z \in \mathcal{X} \setminus S$, and $0 < t \leq q(z)$, let $q' = q + t(\delta_k - \delta_z)$. Then

$$p(x \mid S; q')p(y \mid S; q) = p(y \mid S; q')p(x \mid S; q). \tag{7}$$

Replacing unavailable exposure with exposure to k may change the probability of a choice in $\{x, y\}$. Whenever relative choice is defined at both exposure distributions, the fraction of choices in the pair assigned to x is unchanged. The cross-product formulation accommodates zero choice probabilities. For the same photographer, replace observed purchases of the unavailable lens z with purchases of k. If 60 percent of choices in $\{x, y\}$ were of x, that fraction remains 60 percent whenever the pair has positive choice probability under the new exposure.

The final axiom considers exposure distributions supported entirely on S, so every observed choice is available to the consumer. It requires at least one transfer that raises the recipient's choice probability by less than the transferred frequency.

Axiom 3 (Partial Responsiveness to Exposure)**.** There exist $q \in Q_S$ with $q_S = 1$, distinct $a, b \in S$, and $0 < t \leq q(a)$ such that

$$0 < p\big(b \mid S; q + t(\delta_b - \delta_a)\big) - p(b \mid S; q) < t. \tag{8}$$

The upper bound is imposed at $q_S = 1$, where $q(x) = q(x \mid S)$ for every available alternative.

Figure 3 pairs the exposure transfers with their restrictions on choice. With three available alternatives and one unavailable alternative, exposure lies in a tetrahedron and choice in a triangle.

Theorem 5.1 (Fixed-menu characterization)**.** *Let $S \subsetneq \mathcal{X}$ with $|S| \geq 3$. A choice rule on Q_S satisfies Axioms 1–3 if and only if there exist a unique $\alpha_S \in (0, 1)$ and a unique $p_0(\cdot \mid S) \in \Delta(S)$ such that*

$$p(x \mid S; q) = \frac{(1 - \alpha_S)p_0(x \mid S) + \alpha_S q(x)}{1 - \alpha_S + \alpha_S q_S} \qquad (q \in Q_S,\ x \in S). \tag{9}$$

No continuity or full-support assumption is required on Q_S.

Fix the observed frequency of each unavailable good. Exposure Reallocation Independence makes each available good's choice probability depend only on its own observed frequency. When frequency is transferred between two available goods, the probabilities of all other goods remain fixed. Since choice probabilities sum to one, the two affected probabilities must change by equal and opposite amounts. Applying this requirement across allocations and transfer sizes yields a common linear response. Goods can have different baseline choice probabilities while sharing this response coefficient.

When all observed purchases are available, Partial Responsiveness to Exposure places this common response coefficient strictly between zero and one. Normalizing the remaining fixed component of choice gives the baseline distribution.

Relative Choice Independence connects these affine responses across environments containing unavailable purchases. Replacing an unavailable purchase with an available one preserves relative choice among the other goods at every allocation of their observed frequencies. The intercepts and slopes of their affine choice probabilities must therefore change by a common factor. The requirement that probabilities sum to one determines this factor. These comparisons connect every

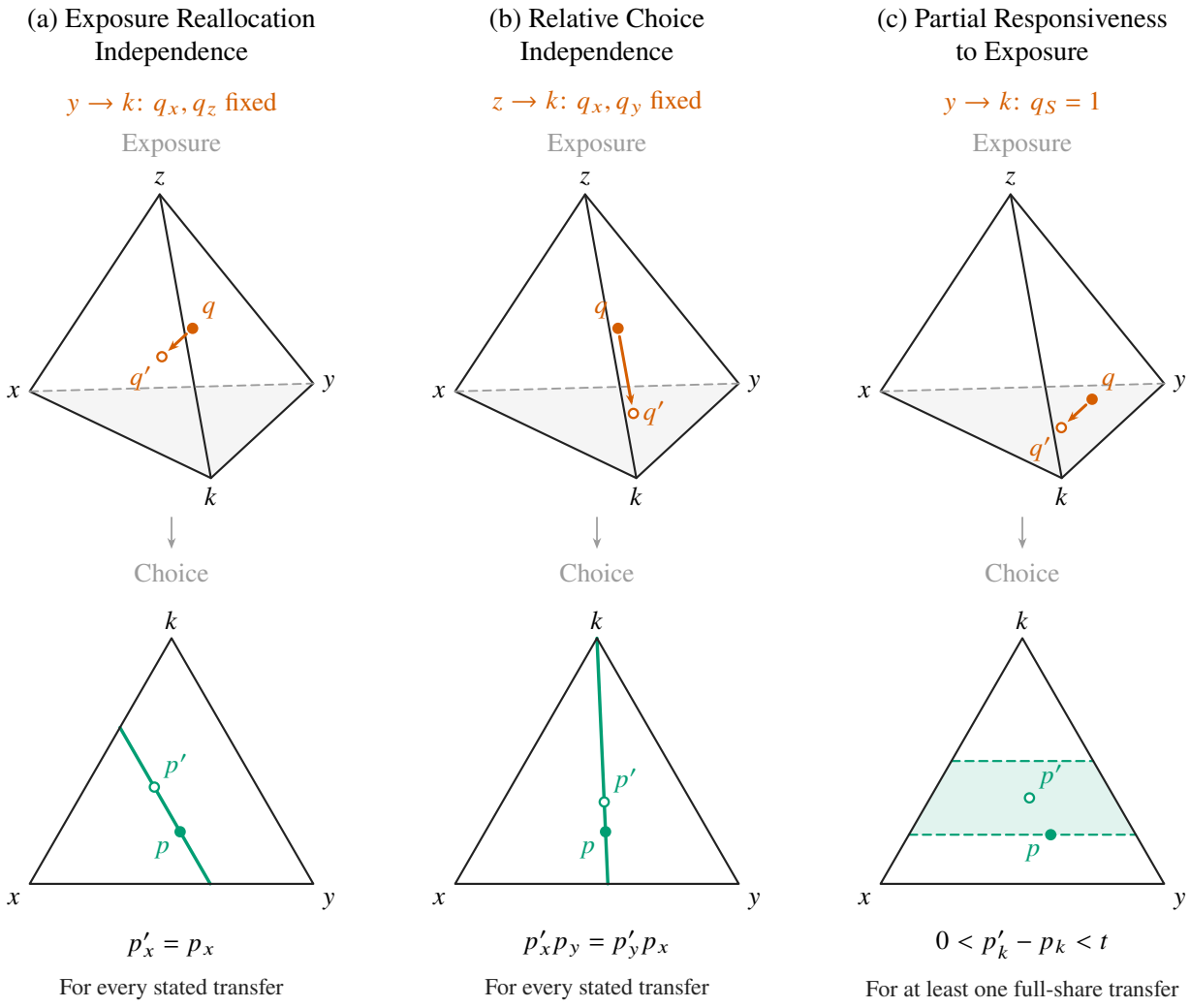


Figure 3: Exposure transfers and choice restrictions. The menu is $S = \{x, y, k\}$ and z is unavailable. Each upper panel shows an exposure transfer from q to q'; the shaded face contains all exposure supported on S. The corresponding lower panel shows the displayed condition on $p' = p(\cdot \mid S; q')$, given the interior point $p = p(\cdot \mid S; q)$. In (a), $p'_x = p_x$, a line parallel to the yk edge. In (b), $p'_x p_y = p'_y p_x$, a line joining vertex k to the xy edge through p. In (c), the open band is $p_k < p'_k < p_k + t$; the dashed boundaries are excluded. The third axiom requires at least one such comparison at $q_S = 1$. Green lines and shading depict each condition separately for the illustrated comparison.

exposure distribution in the domain to the fully available environment, yielding the same baseline throughout. Appendix A.2 gives the proof.

The joint axioms also imply that $q|_S = q'|_S$ entails $p(\cdot \mid S; q) = p(\cdot \mid S; q')$: redistributing observed purchases among unavailable goods leaves choice unchanged. Appendix A.6 illustrates the separate restrictions imposed by the three axioms. Under one-shot fallback, an unavailable draw is followed immediately by a baseline choice. Replacing unavailable exposure with available exposure then reduces the baseline component while leaving direct exposure to the other goods fixed. When these components give different relative frequencies to a pair of those goods, their relative choice changes, violating Relative Choice Independence.

The recovered baseline $p_0(\cdot \mid S)$ is also the limit of choice as q_S tends to zero.[2] The subscript on α_S allows the influence parameter to vary across menus.

5.2 A cross-menu Luce extension

The fixed-menu result allows both the baseline distribution and the influence parameter to vary with the menu. Two comparisons connect these objects across menus. Switching exposure between available alternatives removes the baseline from the change in choice probability. Comparing relative choice between two unexposed alternatives removes their common influence factor.

Let $\mathcal{S}$ be a family of proper menus, each with at least three alternatives, on which the fixed-menu axioms hold. Assume *Full Support*: $p(x \mid S; q) > 0$ for every $S \in \mathcal{S}$, $q \in Q_S$, and $x \in S$. Under Theorem 5.1, this is equivalent to positivity of every baseline coordinate.

Condition C1 (Exposure Response Consistency)**.** For every $S, T \in \mathcal{S}$ and distinct $x, y \in S \cap T$,

$$p(x \mid S; \delta_x) - p(x \mid S; \delta_y) = p(x \mid T; \delta_x) - p(x \mid T; \delta_y). \tag{10}$$

Switching all exposure from y to x produces the same probability increment for x in either menu. Both exposure distributions are fully available on both menus, so the two differences equal α_S and α_T, respectively. The baseline cancels within each menu; only the probability increments are compared.

For the second comparison, keep $x, y \in S$ unexposed and concentrate exposure on $a \in S \setminus \{x, y\}$.

[2] If choice is separately specified at exposure distributions with zero available share, equating it with this limit requires a boundary condition, such as continuity there. See Appendix A.2.

Their probabilities are $(1-\alpha_S)p_0(x \mid S)$ and $(1-\alpha_S)p_0(y \mid S)$. Hence

$$\frac{p(x \mid S;\delta_a)}{p(x \mid S;\delta_a)+p(y \mid S;\delta_a)} = \frac{p_0(x \mid S)}{p_0(x \mid S)+p_0(y \mid S)}. \tag{11}$$

Both members of the pair are available but receive no exposure. Fully available exposure thus reveals their relative baseline choice.

Condition C2 (Baseline Relative Choice Consistency)**.** For every $S, T \in \mathcal{S}$, distinct $x, y \in S \cap T$, $a \in S \setminus \{x, y\}$, and $b \in T \setminus \{x, y\}$,

$$\frac{p(x \mid S;\delta_a)}{p(x \mid S;\delta_a)+p(y \mid S;\delta_a)} = \frac{p(x \mid T;\delta_b)}{p(x \mid T;\delta_b)+p(y \mid T;\delta_b)}. \tag{12}$$

When x and y receive no exposure, their relative choice agrees across menus containing them. Exposure can be concentrated on a different third alternative in each menu. The factors $1-\alpha_S$ and $1-\alpha_T$ cancel within their respective menus, leaving equality of the baseline ratios.

Theorem 5.2 (Cross-menu characterization)**.** *Let $|\mathcal{X}| \geq 4$, and let $\mathcal{S}$ be a family of proper menus, each with at least three alternatives. Suppose every three distinct alternatives in $\mathcal{X}$ lie in some member of $\mathcal{S}$. On each menu, let the rule satisfy Axioms 1–3 and Full Support. Conditions C1 and C2 hold if and only if there exist a common $\alpha \in (0,1)$ and a positive index $u : \mathcal{X} \to \mathbb{R}_{++}$ such that*

$$p(x \mid S;q) = \frac{(1-\alpha)\dfrac{u(x)}{\sum_{y\in S} u(y)} + \alpha q(x)}{1-\alpha+\alpha q_S} \quad (S \in \mathcal{S},\ q \in Q_S,\ x \in S). \tag{13}$$

The influence parameter is unique, and u is unique up to a common positive factor.

This is the Aspiration-Weighted Luce Model. Triple coverage permits baseline ratios within each triple to be read from one menu and connects menu-specific influence parameters through shared pairs. Appendix A.3 proves the result, and Appendix A.6 examines other menu families.

The common index u determines relative baseline choice across menus. Observed relative choice also depends on exposure and the menu. For example, let $\mathcal{X} = \{1,2,3,4,5\}$, $u \equiv 1$, $\alpha = 1/2$, and $q = \delta_1$. The fraction of choices of 1 among choices of 1 or 2 is $4/5$ on $S = \{1,2,3\}$ and $5/6$ on $T = \{1,2,3,4\}$. At the exposure distributions used in C2, conditioning on the pair reveals the common baseline relation.

The representation applies to every observed menu in $\mathcal{S}$ and every positive-share exposure distribution. The fixed-menu identification results below allow menu-specific parameters and zero baseline probabilities.

6 Identification

The identification problem is to separate the consumer's baseline choice from her response to the influencer's observed purchases. Empirical work usually observes only a few exposure regimes. Two regimes recover the baseline and influence parameter when they describe the same consumer in the same choice environment and induce different choice probabilities.

6.1 Regime Stability

On a fixed menu S, write $P = p(\cdot \mid S; q)$ for the choice distribution induced by exposure q, so $P(x) = p(x \mid S; q)$ for each $x \in S$. A *regime* is the pair (q, P). Across regimes, write $P_i = p(\cdot \mid S; q_i)$, where i indexes exposure environments. Identification uses regimes that share the consumer's menu, baseline distribution, and influence parameter.

Assumption 6.1 (Exposure-regime stability)**.** Across the regimes used for identification, the menu S, the baseline distribution $p_0(\cdot \mid S)$, and the influence parameter α remain fixed. A regime affects choice only through the exposure distribution the consumer receives.

Random assignment can isolate a change in exposure. Repeated observations of the same consumer over a short window can make a stable baseline and influence parameter more plausible. Common prices, inventory, geography, and account eligibility help hold the menu fixed. A change in the content supplied to the consumer provides a useful comparison when product quality, prices, and the composition of consumers remain fixed.

The identification result applies when the same (S, p_0, α) governs every observed pair (q_i, P_i). Exposure may depend on variables that leave these objects unchanged. A causal interpretation additionally requires a research design that isolates the effect of exposure.

Under AWI and Assumption 6.1, keeping the observed frequencies of all available goods fixed leaves choice unchanged. Holding the conditional exposure distribution fixed, changing the available share moves choice along the line in Proposition 4.1. Both restrictions can be checked across observed regimes.

6.2 A Single Regime

One exposure regime generally leaves the baseline distribution and influence parameter unidentified. Fix an interior choice distribution P and observed exposure distribution q. For every sufficiently small $\alpha > 0$, the vector

$$p_0^{(\alpha)} = \frac{\big[(1-\alpha) + \alpha q_S\big]P - \alpha q|_S}{1-\alpha}$$

remains a probability distribution and reproduces P. The same observed regime therefore admits a continuum of nearby decompositions. Nonnegativity still bounds the admissible values of α; the failure concerns joint point identification of the influence parameter and the baseline.

Figure 4 shows changes in exposure and consumer choice across two regimes. The baseline distribution and influence parameter must account for both observed choice distributions.

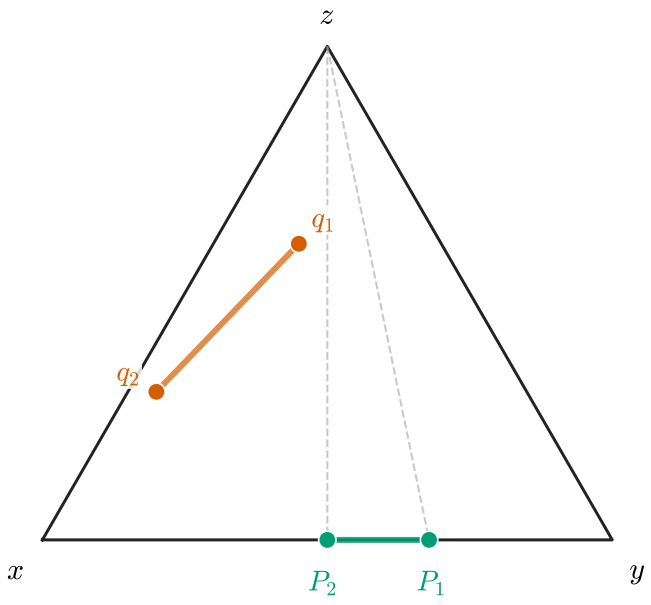


Figure 4: The identification problem. Exposure moves from q_1 to q_2, while choice moves from P_1 to P_2. The hidden baseline and the common influence parameter must account for both movements.

6.3 Two Exposure Regimes

Fix a menu S with $|S| \geq 2$. In regime $i \in \{1, 2\}$, let q_i denote the exposure distribution and P_i the induced choice distribution. Write $v_i = q_i|_S$ for the vector of observed frequencies on S and

$\mu_i = q_{i,S}$ for the available share. Before the final normalization, equation (1) becomes

$$\big[(1-\alpha)+\alpha\mu_i\big]P_i = (1-\alpha)p_0 + \alpha v_i. \tag{14}$$

The factor on the left reverses the normalization in the choice rule. Subtracting the two regime equations eliminates the common baseline.

Define the observed choice movement and the corresponding adjusted exposure movement by

$$\Delta P = P_2 - P_1, \qquad A = (q_2 - q_1)|_S + (1-\mu_2)P_2 - (1-\mu_1)P_1. \tag{15}$$

Rearranging the differenced equation gives the central restriction

$$\Delta P = \alpha A. \tag{16}$$

The vector A combines the change in exposure on the menu with an adjustment for the two regimes' unavailable shares. The observed movement in choice must point in the same direction, and the common scalar is the influence parameter.

Theorem 6.2 (Two-regime identification)**.** *Fix a menu S with $|S| \geq 2$ and two regimes (q_1, P_1) and (q_2, P_2) on that menu, with $v_i = q_i|_S$, $\mu_i = q_{i,S}$, and ΔP and A as in* (15)*. Suppose $A \neq 0$. Compute $\widehat{\alpha}$ from the first formula below; when $\widehat{\alpha} \in (0,1)$, compute $\widehat{p}_0$ from the second:*

$$\widehat{\alpha} = \frac{A^\top \Delta P}{A^\top A}, \qquad \widehat{p}_0 = \frac{\big[(1-\widehat{\alpha})+\widehat{\alpha}\mu_1\big]P_1 - \widehat{\alpha}v_1}{1-\widehat{\alpha}}. \tag{17}$$

Under Assumption 6.1, the two regimes are generated by aspiration-weighted influence if and only if

$$\Delta P = \widehat{\alpha}A, \qquad \widehat{\alpha} \in (0,1), \qquad \widehat{p}_0 \geq 0$$

componentwise. When these conditions hold, $(\widehat{\alpha}, \widehat{p}_0)$ is the unique pair of structural parameters that rationalizes both regimes.

The proportionality in equation (16) asks every coordinate to imply the same influence parameter. Once that restriction holds, equation (17) reveals the baseline; the componentwise inequality checks that the recovered vector can describe choice. When $A = 0$ and $\Delta P \neq 0$, the two regimes reject the model. When $A = \Delta P = 0$, the exposure movement supplies no value for α, although

nonnegativity of the implied baseline can still restrict its admissible range. A reallocation entirely among alternatives outside S falls into this last case: it may be large as a raw exposure change, yet it creates no identifying movement.

The number of testable proportionality restrictions depends on the size of the menu. Both ΔP and A have coordinates summing to zero, so each lies in a space of dimension $|S| - 1$, and proportionality of two such vectors amounts to $|S| - 2$ restrictions. On a binary menu that count is zero: whenever $A \neq 0$, the scalar $\widehat{\alpha}$ reproduces ΔP by construction, and only $\widehat{\alpha} \in (0, 1)$ and $\widehat{p}_0 \geq 0$ retain content. Two regimes on a binary menu therefore recover the parameters under the maintained model. Directional tests require $|S| \geq 3$, and additional regimes supply overidentifying restrictions at any menu size.

The theorem concerns population choice distributions. When A is close to zero, the recovered influence parameter can be sensitive to sampling error in choice and exposure frequencies. When the two regimes share one conditional exposure distribution r and differ only in their available shares,

$$A = \frac{\beta(\mu_2) - \beta(\mu_1)}{\alpha}\,(r - p_0),$$

so the adjusted exposure movement factors into the change in the available share, acting through the weight β, and the distance from the conditional exposure distribution to the baseline distribution. When $r = p_0$, both observed choice distributions equal r and $A = 0$, so changes in the available share leave choice unchanged. When instead the regimes have the same available share μ and different conditional exposure distributions, write $r_i = q_i(\cdot \mid S)$ for the two conditional exposure distributions. Then $A = [\mu + (1-\mu)\beta(\mu)]\,(r_2 - r_1)$, and informativeness depends on the size of the change in r.

Figure 5 makes the argument visible in the three-alternative case $\mathcal{X} = \{x, y, z\}$ with $S = \{x, y\}$. Conditioning any distribution on S moves it along a line through the outside vertex z to the edge xy. The attempt distribution $M_i = M(\cdot \mid S; q_i)$ of Section 3 must therefore lie on the line from z through the observed point P_i. A candidate baseline selects one point on each of these two conditioning lines. The same influence parameter must work in both regimes, so $M_2 - M_1 = \alpha(q_2 - q_1)$: the segment joining the two attempt distributions must be parallel to the observed exposure shift. The baseline in the left panel satisfies this requirement. The candidate in the right panel reaches the observed choices after conditioning, but it implies different influence parameters across the two regimes.

When the two regimes have the same positive available share $\mu_1 = \mu_2 = \mu$, their normalizations

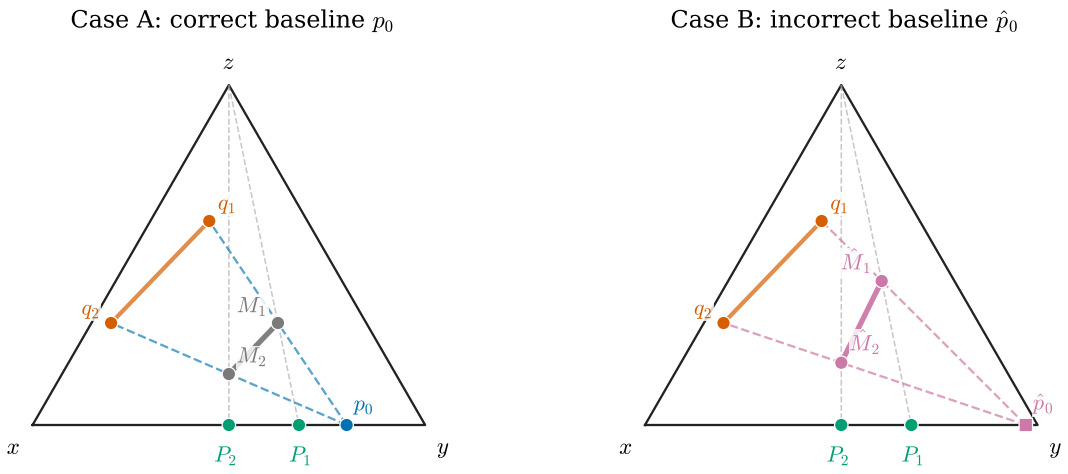


Figure 5: Recovering the hidden baseline. The correct candidate generates attempt distributions M_1 and M_2 whose difference is parallel to the exposure shift. The incorrect candidate fails the common-α restriction.

agree and

$$P_2 - P_1 = \beta(\mu)\big[q_2(\cdot \mid S) - q_1(\cdot \mid S)\big].$$

The coefficient $\beta(\mu)$ is the weight on the conditional exposure distribution in (4). Under the maintained representation, a nonzero change in the conditional exposure distribution at known μ already identifies $\alpha = \beta(\mu)/[\mu + (1-\mu)\beta(\mu)]$; a second share is not required for homogeneous parameter identification.

6.4 Additional Regimes

One informative pair fixes (α, p_0) at the population level. Every additional regime must satisfy equation (1) with those same parameters. Appendix C gives the finite-regime rationalizability conditions and establishes generic identification for designs with several regimes.

These restrictions jointly test the choice rule, stability of (S, p_0, α), and exposure measurement for the same population. Distinguishing among sources of rejection requires evidence about the research design.

With sampled exposure and choice frequencies, the unnormalized equation (14) provides moment conditions for estimation. The identity $\Delta P = \alpha A$ holds in the population, but $\widehat{\Delta P}$ and $\widehat{A}$ are built from the same empirical frequencies, so regressing one on the other correlates the regressor with the error. The unnormalized equation supports minimum-distance estimation with a covari-

ance matrix that accounts for sampling error. Appendix F develops the estimator, specification tests, and confidence sets for weak adjusted exposure movements. It also allows the exposure distribution to be estimated from a separate sample of observations, and treats the case in which changes in the influencer's exposure distribution must be located in a sequence of observations.

When structural parameters are already calibrated and exposure is only partly recorded, the choice distribution restricts the observed frequencies on S to a one-dimensional set. Appendix C derives that set and shows when one additional exposure statistic selects a unique point.

7 The Influencer's Problem

The influencer allocates a fixed amount of content across the products she consumes and displays. Consumer purchases generate product returns, such as promotional commissions, while displays of products unavailable to the consumer can generate value through reputation, exclusivity, or audience growth. Every display corresponds to her actual consumption. She can allocate content freely across the universe $\mathcal{X}$, and q records the relative frequencies of the products displayed.

The consumer's choice rule determines how this allocation affects purchases. The influencer's optimal allocation depends on the influence parameter and the expected product return under the baseline distribution.

Fix a proper menu $S \subsetneq \mathcal{X}$ and an audience whose members share this menu, influence parameter $\alpha \in (0, 1)$, and baseline distribution $p_0(\cdot \mid S) \in \Delta(S)$. Write $\kappa = (1 - \alpha)/\alpha$. The influencer chooses a single exposure distribution $q \in \Delta(\mathcal{X})$, and its available share is $\mu = q_S$. When $\mu > 0$, let $r = q(\cdot \mid S)$ denote the conditional exposure distribution on the menu. The consumer's choice distribution is

$$p(\cdot \mid S; q) = \big[1 - \beta(\mu)\big] p_0(\cdot \mid S) + \beta(\mu) r, \qquad \beta(\mu) = \frac{\mu}{\kappa + \mu}. \tag{18}$$

For this application, assume that the consumer follows the independent repeated sampling procedure in Section 3.2 until an available good is selected. Let τ be the number of attempts required to complete the decision. Its expectation is

$$\mathbb{E}[\tau \mid \mu] = \frac{1}{(1 - \alpha) + \alpha\mu} = \frac{1 + \kappa}{\kappa + \mu}.$$

At $\mu = 0$, each attempt selects the baseline source with probability $1 - \alpha > 0$. Choice is therefore

completed almost surely, with expected completion time $1/(1-\alpha)$.

For each $x \in S$, let $c(x) \geq 0$ be the influencer's product return when the completed decision selects x. If declining to purchase is an available option, it can be represented by an outside option $0 \in S$ with $c(0) = 0$. For each $z \in \mathcal{X} \setminus S$, let $\phi(z) \geq 0$ be the display value of featuring z. This value may capture image, exclusivity, or audience-building benefits that accrue even though the current audience cannot select the displayed product.

All payoffs are measured per initiated decision occasion. The influencer allocates a fixed amount of content, normalized to one, through q. The consumer may require several attempts to choose after observing that content. The expected product return is $c^{\top} p(\cdot \mid S; q)$, while the display value assigned to unavailable products is

$$\sum_{z \in \mathcal{X} \setminus S} \phi(z) q(z).$$

Let $d \geq 0$ be the cost per attempt. The influencer solves

$$\max_{q \in \Delta(\mathcal{X})} \left\{ c^{\top} p(\cdot \mid S; q) + \sum_{z \in \mathcal{X} \setminus S} \phi(z) q(z) - d \frac{1+\kappa}{\kappa + q_S} \right\}. \tag{19}$$

Define

$$c_0 = c^{\top} p_0(\cdot \mid S), \qquad G = \max_{x \in S} c(x) - c_0, \qquad B = \max_{z \in \mathcal{X} \setminus S} \phi(z), \tag{20}$$

and assume $G > 0$ and $B > 0$. The quantity G is the gap between the highest product return on S and the expected product return under the baseline distribution. The quantity B is the highest display value available outside the menu.

Proposition 7.1 (Optimal content allocation)**.** *Fix a proper menu $S \subsetneq \mathcal{X}$, an influence parameter $\alpha \in (0,1)$, a baseline distribution $p_0(\cdot \mid S) \in \Delta(S)$, product returns $c : S \to \mathbb{R}_+$, display values outside the menu $\phi : \mathcal{X} \setminus S \to \mathbb{R}_+$, and a cost per attempt $d \geq 0$. Let $\kappa = (1-\alpha)/\alpha$, define G and B by* (20)*, and suppose $G > 0$ and $B > 0$. Every solution to* (19) *has the same available share*

$$\mu^*(d) = \min \left\{ 1, \max \left\{ 0, \sqrt{\frac{\kappa G + d(1+\kappa)}{B}} - \kappa \right\} \right\}. \tag{21}$$

Whenever $\mu^*(d) > 0$, every optimal conditional exposure distribution on S assigns positive mass only to alternatives maximizing c. Whenever $\mu^*(d) < 1$, every optimal conditional exposure distribution outside S assigns positive mass only to alternatives satisfying $\phi(z) = B$. Hence an optimal

exposure distribution can be chosen with support on at most two alternatives: one available product with the highest product return and one unavailable product with the highest display value.

When completion is costless,

$$\mu^*(0) = \min\left\{1, \max\left\{0, \sqrt{\kappa G/B} - \kappa\right\}\right\}. \tag{22}$$

As $\alpha \to 1$, $\mu^*(0) \to 0$, *whereas for* $d > 0$,

$$\mu^*(d) \longrightarrow \min\left\{1, \sqrt{d/B}\right\}. \tag{23}$$

Proof. Let $\mu = q_S$. When $\mu \in (0, 1)$, write

$$q|_S = \mu r, \qquad q|_{\mathcal{X}\setminus S} = (1-\mu)s,$$

where $r \in \Delta(S)$ and $s \in \Delta(\mathcal{X} \setminus S)$. At $\mu = 0$, the conditional exposure distribution is irrelevant; at $\mu = 1$, the conditional exposure distribution outside the menu is irrelevant. Equation (18) gives

$$c^\top p(\cdot \mid S; q) = c_0 + \beta(\mu)\big(c^\top r - c_0\big).$$

For every $\mu > 0$, $\beta(\mu) > 0$, so an optimal r is supported on $\arg\max_{x\in S} c(x)$. For every $\mu < 1$, the direct display term is

$$(1-\mu)\sum_{z\in\mathcal{X}\setminus S} \phi(z)s(z),$$

so an optimal s is supported on $\arg\max_{z\in\mathcal{X}\setminus S} \phi(z)$. Selecting one maximizer from each set produces an optimal exposure distribution with support on at most two alternatives.

After the conditional distributions r and s have been optimized, the objective equals the constant $c_0 + B$ plus

$$h_d(\mu) = G\frac{\mu}{\kappa+\mu} - B\mu - d\frac{1+\kappa}{\kappa+\mu}.$$

Its derivatives are

$$h_d'(\mu) = \frac{\kappa G + d(1+\kappa)}{(\kappa+\mu)^2} - B, \qquad h_d''(\mu) = -\frac{2[\kappa G + d(1+\kappa)]}{(\kappa+\mu)^3} < 0.$$

Thus h_d is strictly concave, and its unconstrained maximizer is

$$\sqrt{\frac{\kappa G + d(1+\kappa)}{B}} - \kappa.$$

Projection onto $[0, 1]$ gives (21) and proves uniqueness of the available share. Setting $d = 0$ gives (22). Finally, $\alpha \to 1$ is equivalent to $\kappa \downarrow 0$. The costless expression converges to zero, while for $d > 0$ the projected expression converges to $\min\left\{1, \sqrt{d/B}\right\}$, which proves (23). □

Only the scalar c_0 carries the baseline distribution into the influencer's problem. It is the expected product return when no goods available to the consumer are displayed. The optimal conditional distributions r and s do not depend on p_0, and the available share depends on it only through $G = \max_x c(x) - c_0$. The influencer therefore needs the audience's influence parameter and the expected product return c_0 under the baseline distribution.

Conditional on the available share, reallocating exposure among products outside S leaves the consumer's choice distribution unchanged. The influencer nevertheless assigns that exposure to products with the highest display value. The identity of an unavailable product can therefore affect the influencer's optimal display allocation while leaving consumer choice unchanged at a fixed available share and conditional exposure distribution.

Raising μ increases expected product return through $G\mu/(\kappa+\mu)$, sacrifices $B\mu$ units of display value, and shortens expected completion. The gain in expected product return and the reduction in expected delay cost are concave in the available share. The loss of display value is linear, yielding the unique optimum in Proposition 7.1.

When $d = 0$, the optimal available share is weakly single-peaked in the influence parameter. It becomes positive once $\alpha > B/(B+G)$. Before projection onto $[0, 1]$, write

$$\widetilde{\mu}(\kappa) = \sqrt{\kappa G/B} - \kappa.$$

Since

$$\widetilde{\mu}'(\kappa) = \frac{1}{2}\sqrt{\frac{G}{B\kappa}} - 1,$$

the unconstrained share reaches its maximum at $\kappa = G/(4B)$, or equivalently at

$$\alpha = \frac{4B}{G+4B}.$$

Projection preserves weak single-peakedness. If $G/B > 4$, the upper bound creates an interval on which the influencer displays only goods available to the audience.

At high influence parameters, a small available share can already induce a strong choice response. Allocating more exposure to unavailable goods then provides greater display value. Let $V^*(\alpha)$ denote the maximized value of (19) when $d = 0$. At an interior solution,

$$V^*(\alpha) = c_0 + B + \left(\sqrt{G} - \sqrt{\kappa B}\right)^2.$$

This value rises strictly as α rises. At the lower corner, $V^*(\alpha) = c_0 + B$; at the upper corner, $V^*(\alpha) = c_0 + \alpha G$. The maximized payoff is therefore weakly increasing over the entire parameter range and strictly increasing whenever the optimal available share is positive. On the right side of Figure 6, a more responsive audience gives the influencer a higher payoff while requiring a smaller available share.

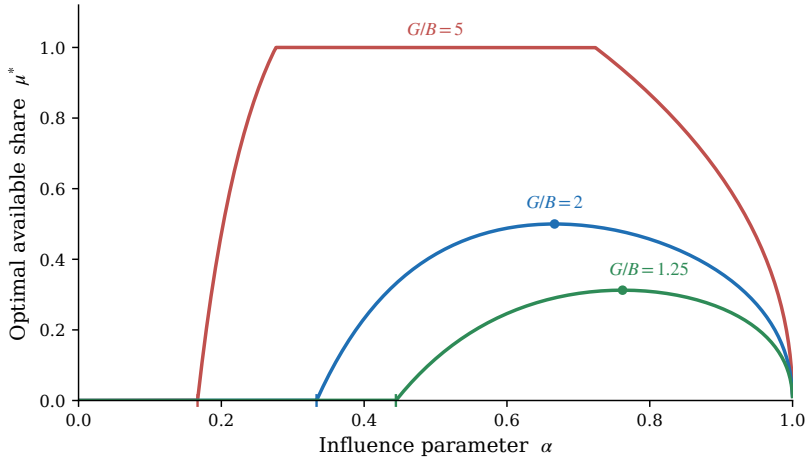


Figure 6: The influencer's optimal available share when completion is costless, for three values of G/B.

At an interior optimum with $d = 0$,

$$\mathbb{E}[\tau \mid \mu^*(0)] = \frac{1 + \kappa}{\sqrt{\kappa G/B}}.$$

As $\alpha \to 1$, expected completion time grows at rate $\kappa^{-1/2}$. Under the repeated sampling procedure adopted here, the declining available share increases the expected time to complete a choice. A positive cost per attempt produces the positive high-influence limit in (23).

The exposure distribution allocates the fixed amount of content. Display value is earned from

that allocation, while the consumer's repeated sampling determines the time required to complete a choice. If display value instead accrued on every unavailable draw, the expected number of such draws would be $(1-\mu)/(\kappa+\mu)$. After optimizing the conditional distributions r and s, the payoff would become

$$\widetilde{V}_d(\mu) = c_0 + \frac{(G-B)\mu + B - d(1+\kappa)}{\kappa+\mu},$$

with derivative

$$\widetilde{V}_d'(\mu) = \frac{\kappa G + (1+\kappa)(d-B)}{(\kappa+\mu)^2}.$$

The derivative has a constant sign over μ. Assigning display value to each unavailable draw therefore yields $\mu^* = 1$ when $\kappa G + (1+\kappa)(d-B) > 0$ and $\mu^* = 0$ when this expression is negative; every share is optimal at equality. The solution is generically a corner, so the interior allocation and the single-peaked available-share curve disappear. In (19), display value instead accrues to the fixed amount of content, and repeated sampling determines expected delay cost.

Remark 7.2 (Display values for available goods). Suppose ϕ is defined on all of $\mathcal{X}$, so available products carry direct display value as well as product return. The direct display term in (19) becomes $\sum_{x\in\mathcal{X}} \phi(x)q(x)$. For a fixed $\mu \in (0,1]$, the terms that depend on the conditional exposure distribution are

$$\beta(\mu)\sum_{x\in S} c(x)r(x) + \mu\sum_{x\in S}\phi(x)r(x) = \mu\sum_{x\in S}\left[\phi(x) + \frac{c(x)}{\kappa+\mu}\right]r(x).$$

An optimal conditional exposure distribution can therefore be concentrated on an alternative

$$x_\mu \in \arg\max_{x\in S}\left\{\phi(x) + \frac{c(x)}{\kappa+\mu}\right\}.$$

Whenever $\mu < 1$, unavailable exposure can be concentrated on an alternative maximizing $\phi(z)$ over $z \in \mathcal{X}\setminus S$. Recalling B from (20) as the maximum over $\mathcal{X}\setminus S$, the remaining choice is the one-dimensional problem

$$\max_{\mu\in[0,1]}\left\{c_0 + (1-\mu)B - d\frac{1+\kappa}{\kappa+\mu} + \mu\max_{x\in S}\left[\phi(x) + \frac{c(x)-c_0}{\kappa+\mu}\right]\right\}.$$

An optimal exposure distribution can again be chosen with support on at most two alternatives. The available product may change with μ because display value and product return receive different relative weights as the available share changes. At $\mu = 0$, the conditional exposure distribution is

irrelevant. Setting $\phi(x) = 0$ for every $x \in S$ recovers Proposition 7.1 and its closed-form solution.

8 Conclusion

An influencer's choices can affect a consumer who has access to only some of the same goods. The model makes the consumer's choice probabilities a weighted average of the baseline distribution and the conditional exposure distribution. The weight on the conditional exposure distribution rises with the available share. A given change in the relative frequencies among available goods therefore produces a larger response at a higher available share. The identities of the unavailable goods have no further effect on choice.

For menus with at least three available alternatives and at least one unavailable alternative, three restrictions characterize this choice rule and recover a unique baseline and influence parameter. Under the model, two regimes producing different consumer choice distributions identify both objects when the menu and parameters remain fixed. Additional cross-menu restrictions yield a common influence parameter and a Luce baseline distribution. The identifying data include both the available share and the influencer's relative purchase frequencies within the consumer's menu.

Product returns favor displays of goods available to the consumer, while display value can support displays of unavailable goods. Under the repeated sampling interpretation, a low available share also delays choice. A positive cost per attempt weakly increases the optimal available share and keeps it bounded away from zero as the influence parameter approaches one.

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A Proofs

A.1 Proof of Proposition 4.1

Proof. Let $r = q(\cdot \mid S) = q'(\cdot \mid S)$. Equation (4) gives

$$p(\cdot \mid S; q) = [1 - \beta(q_S)]p_0 + \beta(q_S)r$$

and the analogous expression for q'. Subtracting yields (5). Moreover,

$$\beta'(\mu) = \frac{\alpha(1-\alpha)}{[(1-\alpha)+\alpha\mu]^2} > 0.$$

Hence $q'_S < q_S$ implies $0 < \beta(q'_S) < \beta(q_S)$. Both choice distributions lie on the segment from p_0 to r. Choice under q' lies between p_0 and choice under q. □

A.2 Proof of Theorem 5.1

Fix a proper menu $S \subsetneq \mathcal{X}$ with $|S| \geq 3$. In this proof, w denotes the entire vector $q|_{\mathcal{X}\setminus S}$ and ranges over $\mathcal{W}_S = \{w \in \mathbb{R}_+^{\mathcal{X}\setminus S} : \sum_{z\notin S} w_z < 1\}$. Write $\mu(w) = 1 - \sum_{z\notin S} w_z$. When a unit vector δ_x appears in a coordinate space indexed by a subset of $\mathcal{X}$ containing x, it denotes the restriction of that vector to the indicated space.

Lemma A.1 (Response conditional on unavailable exposure)**.** *Exposure Reallocation Independence holds if and only if there exist unique coefficient functions $a_x : \mathcal{W}_S \to \mathbb{R}$ for $x \in S$ and $h : \mathcal{W}_S \to \mathbb{R}$ such that*

$$p(x \mid S; q) = a_x(w) + h(w)q(x), \qquad x \in S, \tag{24}$$

where $w = q|_{\mathcal{X}\setminus S}$, and, at each w,

$$\sum_{x\in S} a_x(w) = 1 - \mu(w)h(w), \qquad a_x(w) \geq 0, \quad a_x(w) + \mu(w)h(w) \geq 0 \quad (x \in S). \tag{25}$$

The axiom imposes no additional restriction across different vectors w; in particular, the slope need not be positive.

Proof. Fix w and abbreviate $\mu = \mu(w) > 0$. The vector $v = q|_S$ ranges over all nonnegative vectors summing to μ. For fixed v_x, any two allocations of the other coordinates can be connected by finitely many feasible transfers among $S \setminus \{x\}$: move mass from a coordinate above its target to one below its target. The axiom preserves choice of x at every step. Thus $p(x \mid S; q) = f_x(v_x)$ for a function $f_x : [0, \mu] \to [0, 1]$, and $\sum_{x\in S} f_x(v_x) = 1$ whenever $v \geq 0$ sums to μ.

Put $a_x = f_x(0)$ and $g_x(s) = f_x(s) - a_x$. For distinct x, y, choose $k \in S \setminus \{x, y\}$. Applying the adding-up equality to $s\delta_x + (\mu - s)\delta_k$ and $s\delta_y + (\mu - s)\delta_k$ gives $g_x(s) = g_y(s)$. Denote the common increment function by g. Applying the same equality to $s\delta_x + t\delta_y + (\mu - s - t)\delta_k$ and $(s + t)\delta_x + (\mu - s - t)\delta_k$ gives $g(s + t) = g(s) + g(t)$ for $s, t \geq 0$ with $s + t \leq \mu$. Also $|g(s)| \leq 1$ because each f_x takes values in $[0, 1]$.

For every positive integer n and $0 \leq s \leq \mu/n$, $|g(s)| = |g(ns)|/n \leq 1/n$. Hence g is continuous at zero. Additivity gives continuity elsewhere, and rational approximation yields $g(s) = hs$ for all $s \in [0, \mu]$. This proves (24) without assuming regularity of the observed rule. Adding up gives the first part of (25); evaluating v_x at zero and at μ gives the inequalities. Conversely, these inequalities make each affine coordinate nonnegative throughout $[0, \mu]$, and adding up makes the

vector a distribution satisfying the axiom. Its values at zero and positive own exposure uniquely determine each intercept and the common slope. □

Proof of Theorem 5.1. Apply Lemma A.1. At $w = 0$, the available share is one and an exposure transfer of size t changes its recipient's probability by $h(0)t$. Partial Responsiveness to Exposure therefore gives $\alpha_S := h(0) \in (0, 1)$. Also $a_x(0) \geq 0$ and $\sum_{x \in S} a_x(0) = 1 - \alpha_S$.

Fix w, an unavailable coordinate z with $w_z > 0$, and $d \in (0, w_z]$. Let $w' = w - d\delta_z$, $\mu = \mu(w)$, and $\mu' = \mu + d$; use primes for the coefficients at w'. For any $x \in S$, choose distinct $y, k \in S \setminus \{x\}$. Give x exposure zero, y exposure s, and k exposure $\mu - s$, for $s \in [0, \mu]$. Transfer d from z to k. Relative Choice Independence gives $a_x(a'_y + h's) = (a_y + hs)a'_x$. Subtracting the equalities at $s = 0$ and $s = \mu > 0$ gives $a_x h' = ha'_x$. Summing over x and using (25) gives $(1 - \mu h)h' = h(1 - \mu' h')$. Hence

$$a_x h' = ha'_x \quad (x \in S), \qquad h - h' = dhh'. \tag{26}$$

These identities remain valid when a choice probability or slope is zero.

Define $\kappa_S = (1 - \alpha_S)/\alpha_S > 0$ and $p_0(x \mid S) = a_x(0)/(1 - \alpha_S)$. This baseline lies in $\Delta(S)$. At $w = 0$, $h(0) = 1/(\kappa_S + 1)$ and $a_x(0) = \kappa_S p_0(x \mid S)/(\kappa_S + 1)$. From any w, remove its positive coordinates one at a time to obtain a finite path to zero, and work backwards along this path. If at its next point $h' = 1/(\kappa_S + \mu')$ and $a'_x = \kappa_S p_0(x \mid S)/(\kappa_S + \mu')$, then (26) gives $h(1 - dh') = h'$, with $1 - dh' = (\kappa_S + \mu)/(\kappa_S + \mu') > 0$. Thus $h = 1/(\kappa_S + \mu) > 0$, and the first identity gives $a_x = \kappa_S p_0(x \mid S)/(\kappa_S + \mu)$. Induction establishes these formulas at every w. Both coefficients depend on w only through the available share $\mu(w)$. Substituting them in (24) yields (9).

Conversely, a within-menu transfer in (9) leaves the denominator and all unaltered numerators fixed. A transfer from an unavailable alternative to k changes the common denominator and only k's numerator, so (7) holds by cross multiplication. At an available share of one the recipient gains $\alpha_S t$; taking $q = \delta_a$, $b \neq a$, and $t = 1$ verifies the existential response condition. Finally, the unique full-share slope $h(0)$ identifies α_S, and the full-share intercepts identify $p_0(\cdot \mid S)$ uniquely. □

Zero-share choices. If $q_{n,S} \to 0$, each $q_n(x)$ for $x \in S$ tends to zero, so the represented choice tends to $p_0(\cdot \mid S)$. For a separately specified zero-share exposure distribution q^0, the exposure distributions $q_n = (1 - 1/n)q^0 + (1/n)\delta_x$, $x \in S$, converge to it from Q_S. Continuity at q^0 therefore forces its choice to be $p_0(\cdot \mid S)$. Assigning that value at every exposure distribution with zero available share gives the unique extension continuous there; the positive-domain axioms alone leave

those actual values unrestricted.

A.3 Proof of Theorem 5.2

Lemma A.2 (The observable cross-menu comparisons)**.** *Suppose the fixed-menu representation holds on every observed menu. For distinct $x, y \in S$,*

$$p(x \mid S; \delta_x) - p(x \mid S; \delta_y) = \alpha_S. \tag{27}$$

Under Full Support, for $a \in S \setminus \{x, y\}$,

$$\frac{p(x \mid S; \delta_a)}{p(x \mid S; \delta_a) + p(y \mid S; \delta_a)} = \frac{p_0(x \mid S)}{p_0(x \mid S) + p_0(y \mid S)}. \tag{28}$$

Consequently, Condition C1 equates influence parameters of menus sharing a pair, and Condition C2 equates their baseline ratios for that pair, without requiring common influence.

Proof. Each displayed exposure distribution is supported on the corresponding menu, so the denominator in (9) is one. Subtraction gives (27). In (28), x and y are unexposed, and their choice probabilities are $(1-\alpha_S)p_0(x \mid S)$ and $(1-\alpha_S)p_0(y \mid S)$. The common positive factor cancels. For positive coordinates, equality of these conditional frequencies is equivalent by cross multiplication to equality of the ratios of the baseline coordinates. □

Proof of Theorem 5.2. If a common (α, u) gives (13), each baseline is strictly positive. The fixed-menu theorem gives the three axioms, and baseline positivity gives Full Support. Lemma A.2 gives both cross-menu conditions, proving necessity.

Conversely, the fixed-menu theorem recovers $(\alpha_S, p_0(\cdot \mid S))$ on each menu. Full Support makes every recovered baseline coordinate positive: a zero coordinate at x would give $p(x \mid S; \delta_y) = 0$ for $y \neq x$ in S.

Every pair $\{x, y\}$ lies in an observed menu, because it can be extended to a triple. By Condition C1 and (27), all menus containing that pair have the same influence parameter; denote this temporary value by α_{xy}. A menu covering a triple makes all three of its pair values equal. Two pairs with a common alternative therefore have equal values. Disjoint pairs $\{x, y\}$ and $\{a, b\}$ are connected through $\{x, a\}$, using the triples $\{x, y, a\}$ and $\{x, a, b\}$. Hence every pair value, and thus every menu's influence parameter, equals one $\alpha \in (0, 1)$.

For distinct x, y, define the positive ratio $\omega(x, y) = p_0(x \mid S)/p_0(y \mid S)$ using any menu containing the pair, and set $\omega(x, x) = 1$. Condition C2 and Lemma A.2 make this definition independent of the selected menu. For distinct x, y, k, read the ratios from a menu containing all three to obtain

$$\omega(x, y)\omega(y, k) = \omega(x, k). \tag{29}$$

The identity extends to repeated indices because $\omega(y, x) = 1/\omega(x, y)$. Fix $x_0 \in \mathcal{X}$, normalize $u(x_0) = 1$, and set $u(x) = \omega(x, x_0)$ for $x \neq x_0$. Then (29) gives $\omega(x, y) = u(x)/u(y)$. On each menu, the baseline has the same ratios as u and sums to one, so $p_0(x \mid S) = u(x)/\sum_{y \in S} u(y)$. Substitution yields (13).

The response difference (27) identifies α. Every pair is observed, so any other positive index generating the family has all the same ratios and differs from u by one common positive factor. This proves uniqueness. □

A.4 Proof of Theorem 6.2

Proof. Under the model, equation (14) holds in both regimes. Subtracting regime 1 from regime 2 and collecting terms gives

$$(1 - \alpha)(P_2 - P_1) = \alpha\big[(v_2 - v_1) - \mu_2 P_2 + \mu_1 P_1\big].$$

Adding $\alpha(P_2 - P_1)$ to both sides yields equation (16). When $A \neq 0$, taking its inner product with A shows that any rationalizing influence parameter must equal $\widehat{\alpha}$ in equation (17). Rearranging the regime-1 equation then gives the stated expression for $\widehat{p}_0$. Necessity of all three conditions follows.

For sufficiency, suppose the conditions in the theorem hold. Reversing the algebra that leads to equation (16) gives

$$\big[(1 - \widehat{\alpha}) + \widehat{\alpha}\mu_2\big]P_2 - \widehat{\alpha}v_2 = \big[(1 - \widehat{\alpha}) + \widehat{\alpha}\mu_1\big]P_1 - \widehat{\alpha}v_1.$$

By equation (17), the right side equals $(1 - \widehat{\alpha})\widehat{p}_0$, so the same pre-normalization equation holds in both regimes. Moreover,

$$(1 - \widehat{\alpha})\mathbf{1}^{\mathsf{T}}\widehat{p}_0 = (1 - \widehat{\alpha}) + \widehat{\alpha}\mu_1 - \widehat{\alpha}\mu_1 = 1 - \widehat{\alpha}.$$

Thus $\widehat{p}_0$ sums to one. Componentwise nonnegativity places it in $\Delta(S)$, and equation (1) then reproduces both choice distributions. Finally, $A \neq 0$ makes $\widehat{\alpha}$ the only scalar satisfying equation (16); the regime equation then makes $\widehat{p}_0$ unique. □

A.5 Proof of Proposition 7.1

The proof, including the case of a positive cost per attempt, is given immediately after Proposition 7.1 in Section 7.

A.6 Examples and scope of the characterization

Separate content of the fixed-menu axioms. A constant rule $p(\cdot \mid S; q) = p_0$ satisfies both independence axioms but not Partial Responsiveness to Exposure. The rule $p(x \mid S; q) = e^{q(x)}/\sum_{y\in S} e^{q(y)}$ satisfies Relative Choice Independence: the compared probabilities share a denominator, and their numerators are unchanged. The transfer $\delta_a \to \delta_b$ at full share increases the recipient's probability by $(e-1)/(e+|S|-1) \in (0,1)$, so partial responsiveness also holds. However, for distinct $a, b, k \in S$, compare $q = (\delta_a + \delta_b)/2$ with $q' = q + t(\delta_b - \delta_a)$, $0 < t < 1/2$. The numerator for k is unchanged while the denominator increases, because $e^{1/2-t} + e^{1/2+t} > 2e^{1/2}$. Choice of k decreases, violating Exposure Reallocation Independence.

The one-shot fallback rule provides the third example. Take any $|S| = m \geq 3$, a uniform baseline distribution, and $\alpha = 1/2$. The rule $p(\cdot \mid S; q) = (1 - \alpha q_S)p_0 + \alpha q|_S$ satisfies Exposure Reallocation Independence and Partial Responsiveness to Exposure. For distinct $x, y, k \in S$ and $z \notin S$, let $q = \frac{1}{4}\delta_x + \frac{1}{4}\delta_k + \frac{1}{2}\delta_z$ and $q' = q + \frac{1}{4}(\delta_k - \delta_z)$. Relative choice of x within $\{x, y\}$ changes from $(m+6)/(m+12)$ to $(m+5)/(m+10)$; these differ for every $m \geq 3$. At $m = 3$ they are $3/5$ and $8/13$. Thus Relative Choice Independence fails. These examples establish the separate content of the three axioms on every fixed proper menu of the stated size.

The response to raw exposure frequencies. Allow the coefficient on raw exposure frequency to vary across available alternatives. Consider

$$p^{\eta}(x \mid S; q) = \frac{\kappa p_0(x \mid S) + \eta_x q(x)}{\kappa + \sum_{y\in S} \eta_y q(y)}, \qquad \kappa > 0, \quad p_0(\cdot \mid S) \in \Delta(S), \quad \eta_x > 0.$$

This rule satisfies Relative Choice Independence. Transferring exposure of size t from available alternative a to available alternative b changes the denominator by $t(\eta_b - \eta_a)$. If $\eta_a \neq \eta_b$, the choice probability of every other alternative with a positive numerator changes even though its exposure remains fixed.

Zero probabilities and binary menus. If both pair totals are positive, the cross-product equality in Relative Choice Independence is equivalent to equality of relative choice. If both pair coordinates vanish at one exposure distribution, that equality alone says nothing about the pair at the other. The fixed-menu proof compares coefficients over the whole exposure domain, so its argument also covers zero choice probabilities. For $|S| = 2$, both independence axioms are vacuous. To see that partial responsiveness alone does not restore the representation, let $S = \{x, y\}$, $r = q(x)/q_S$, and set $p(x \mid S; q) = 1/4 + r/4 + r^2/4$ and $p(y \mid S; q) = 1 - p(x \mid S; q)$. At full share, the transfer $\delta_y \to \delta_x$ gives a probability increment of $1/2$, so the response condition holds. The rule is nonlinear in r on a fixed-share interval and therefore has no AWI representation. This example does not affect binary-menu identification under a maintained AWI rule.

Why the cross-menu condition concerns the baseline. Under the common representation, let $U_S = \sum_{k \in S} u(k)$ and $\kappa = (1 - \alpha)/\alpha$. For $x, y \in S$,

$$\frac{p(x \mid S; q)}{p(y \mid S; q)} = \frac{\kappa u(x) + U_S q(x)}{\kappa u(y) + U_S q(y)}.$$

Changing the menu changes U_S, so this ratio need not be menu-independent. Taking $q(x) = q(y) = 0$ removes that dependence and recovers the baseline ratio. For the example in the main text, the full choice vectors are $(2/3, 1/6, 1/6)$ on $\{1, 2, 3\}$ and $(5/8, 1/8, 1/8, 1/8)$ on $\{1, 2, 3, 4\}$. Their pair-conditional frequencies for option 1 within $\{1, 2\}$ are $4/5$ and $5/6$.

Pairwise agreement need not give a common index. Let $\mathcal{X} = \{1, \ldots, 6\}$ and take $S_1 = \{1, 2, 3, 4\}$, $S_2 = \{3, 4, 5, 6\}$, and $S_3 = \{5, 6, 1, 2\}$. Assign the baseline $(2, 2, 1, 1)/6$ in the displayed order on each menu, and use one common influence parameter. Each shared pair has relative baseline choice $1/2$ in both menus, so both cross-menu conditions hold. A common index would nevertheless require $u(1)/u(3) = u(3)/u(5) = u(5)/u(1) = 2$, whose product is 8, violating the required product of 1. No positive index satisfies these requirements. This family covers every pair and omits some triples.

What weaker menu overlap identifies. When existence of a common positive index is maintained, connect menus that share at least one alternative. Their within-menu ratios fix their relative normalizations along each such connection. A connected family therefore identifies the index up to one factor on its observed union. Disconnected components retain separate factors, and alternatives never observed remain unidentified. Without maintained existence, ratios obtained along different paths must also agree. Condition C1 instead uses a pair shared by two menus. On a weaker domain it equates influence parameters only along chains of menus with intersections of size at least two. Single-alternative overlap cannot replace that requirement for this condition. Triple coverage in the theorem supplies both kinds of connection.

Exposure domain. The axioms impose finite-transfer equalities throughout $\mathcal{Q}_S$; agreement at selected exposure distributions does not verify them. On the full menu Axiom 2 is vacuous; on a singleton, Axiom 3 does not apply. The latter's upper bound concerns only $q_S = 1$.

B Sampling Procedures and Choice Probabilities

Consider a fixed consumer who draws once from the mixture of her baseline and exposure distributions. If an unavailable draw ends the decision without a choice, the distribution conditional on completion is AWI. Persistent redrawing from the same mixture produces this distribution as well. Returning to the baseline after an unavailable draw generally changes that distribution.

Fix the consumer's menu S, baseline distribution $p_0(\cdot \mid S)$, and influence parameter $\alpha \in (0, 1)$. For $q_S > 0$, write $\mu = q_S$ and $r = q(\cdot \mid S)$. The choice distribution generated by each procedure in Table 1 can be written as $p = [1 - b(\mu)]p_0 + b(\mu)r$, where $b(\mu)$ is the weight on the conditional exposure distribution. At a given available share, this weight determines the response of choice probabilities to changes in r.

For the finite-cap rule, selecting an available alternative from the exposure distribution on attempt t has probability $[\alpha(1-\mu)]^{t-1}\alpha\mu$. Every path that remains after attempt N ends with baseline choice. The resulting weight on the conditional exposure distribution is

$$b_N(\mu) = \alpha\mu \sum_{t=0}^{N-1} [\alpha(1-\mu)]^t = \frac{\alpha\mu}{1-\alpha+\alpha\mu}\left[1 - \{\alpha(1-\mu)\}^N\right]. \tag{30}$$

As N tends to infinity, $b_N(\mu)$ converges to the persistent-redrawing weight in the main text.

Sampling procedure	Treatment of unavailable alternatives	Weight $b(\mu)$
Filter first	Remove exposure to unavailable alternatives before selecting a source	α
One-shot fallback	After drawing an unavailable alternative, choose from the baseline distribution	$\alpha\mu$
At most N attempts	Redraw after an unavailable alternative; if all N attempts fail, choose from the baseline distribution	$b_N(\mu)$ in (30)
Persistent redrawing	Redraw until an available alternative is selected	$\dfrac{\alpha\mu}{1-\alpha+\alpha\mu}$

Table 1: Weights on the conditional exposure distribution under four sampling procedures.

The capped rule needs a convention for occasions on which all N attempts land outside the menu. The rule defining b_N assigns each such occasion to a baseline choice, so every occasion ends in a completed choice. Under an abandonment convention these occasions instead remain incomplete. The distribution conditional on completion then places the following weight on the conditional exposure distribution:

$$\frac{b_N(\mu)}{1-[\alpha(1-\mu)]^N} = \frac{\alpha\mu}{1-\alpha+\alpha\mu}$$

for every $N \geq 1$. Thus capped abandonment and persistent redrawing agree on the distribution of completed choices. Choosing an available no-purchase option in S completes the decision. Conditioning on purchases further excludes that option and renormalizes the probabilities. The procedures can differ in completion probabilities and durations; conditioning on completion can also change type weights in pooled samples.

On a proper menu $S \subsetneq \mathcal{X}$ with $|S| \geq 3$, all four rules satisfy Exposure Reallocation Independence and Partial Responsiveness to Exposure on the positive-share domain. They also leave the allocation of unavailable exposure irrelevant. Among the four choice rules in Table 1, with baseline choice after an exhausted finite cap, Relative Choice Independence distinguishes the AWI response weight. At an available share of one, each rule raises the recipient's choice probability by αt for a transfer of size t between available alternatives. Theorem 5.1 then determines the weight on the conditional exposure distribution at every available share:

$$b(\mu) = \frac{\alpha\mu}{1-\alpha+\alpha\mu}.$$

Filter-first, one-shot fallback, and every fixed finite retry cap differ from this weight at some positive available share and therefore cannot satisfy Relative Choice Independence on the whole exposure domain.

Implications for the influencer's problem. Consider the influencer's problem with zero cost per attempt. Retain $G > 0$ and $B > 0$ from (20): G is the gain over the baseline expected product return, and B is the highest display value. Under the finite-cap rule, let $\mu_N^*(\alpha)$ maximize $Gb_N(\mu) - B\mu$ over $[0, 1]$.

Proposition B.1 (High-influence limit under a finite retry cap)**.** *Fix a finite $N \geq 1$ and constants $G > 0$ and $B > 0$.*

(i) *As $\alpha \to 1$, $b_N(\mu)$ converges uniformly on $[0, 1]$ to $1 - (1 - \mu)^N$.*

(ii) *If $N = 1$, then $\mu_1^*(\alpha) \to 1$ when $G > B$ and $\mu_1^*(\alpha) \to 0$ when $G \leq B$.*

(iii) *If $N \geq 2$, then*

$$\mu_N^*(\alpha) \longrightarrow \begin{cases} 0, & GN \leq B, \\ 1 - \left(\dfrac{B}{GN}\right)^{1/(N-1)}, & GN > B. \end{cases}$$

Proof. The expression in (30) is continuous on the compact square $[0, 1]^2$ after setting its value at $\alpha = 1$ by continuity. At $\alpha = 1$,

$$b_N(\mu) = \mu \sum_{t=0}^{N-1} (1 - \mu)^t = 1 - (1 - \mu)^N.$$

Uniform continuity gives uniform convergence as $\alpha \to 1$. The influencer objectives therefore converge uniformly to

$$h_N^\infty(\mu) = G[1 - (1 - \mu)^N] - B\mu.$$

For $N = 1$, the finite-α objective is $(\alpha G - B)\mu$, which gives part (ii). For $N \geq 2$, h_N^∞ is strictly concave and

$$(h_N^\infty)'(\mu) = GN(1 - \mu)^{N-1} - B.$$

Its unique maximizer is the value in part (iii). Uniform convergence to an objective with a unique maximizer implies convergence of the finite-α maximizers. □

With zero cost per attempt, persistent redrawing sends the optimal available share toward zero as

α approaches one. Under a fixed finite cap, the limiting share is positive when $GN > B$. The high-influence behavior in Figure 6 therefore depends on how the weight on the conditional exposure distribution responds to the available share.

C Further Identification Results

C.1 One and Many Regimes

A single exposure regime generally leaves both the baseline distribution and the influence parameter unidentified.

Proposition C.1 (Single-regime nonidentification)**.** *Fix a menu S, an exposure distribution $q \in \Delta(X)$, and an interior choice distribution $P \in \Delta^\circ(S)$. There is an interval of influence parameters $\alpha > 0$ and associated full-support baselines $p_0^{(\alpha)}$ for which aspiration-weighted influence maps q to P.*

Proof. For $\alpha \in (0, 1)$ define

$$p_0^{(\alpha)} = \frac{[(1-\alpha) + \alpha q_S]P - \alpha q|_S}{1-\alpha}.$$

The vector sums to one, and substitution into (1) reproduces P. As $\alpha \downarrow 0$, $p_0^{(\alpha)} \to P$ coordinatewise. Since P has full support, $p_0^{(\alpha)}$ remains strictly positive for every sufficiently small $\alpha > 0$. □

Fix a menu S and $K \geq 2$ regimes. In each regime i, let q_i be the exposure distribution and P_i the consumer's choice distribution on S. Define $\Delta_{ij} = P_j - P_i$ and

$$A_{ij} = (q_j - q_i)|_S + (1 - q_{j,S})P_j - (1 - q_{i,S})P_i.$$

Also define $D_i(\alpha) = (1-\alpha) + \alpha q_{i,S}$ and $\psi_i(\alpha) = D_i(\alpha)P_i - \alpha q_i|_S$.

Proposition C.2 (Finite-regime rationalizability)**.** *Let*

$$\mathcal{A} = \left\{\alpha \in (0,1) : \Delta_{ij} = \alpha A_{ij} \text{ for every } i < j,\ \psi_i(\alpha) \in \mathbb{R}^S_+ \text{ for some } i\right\}.$$

The K regimes are generated by common parameters (α, p_0) if and only if $\mathcal{A}$ is nonempty. For each $\alpha \in \mathcal{A}$, the pairwise restrictions imply $\psi_1(\alpha) = \cdots = \psi_K(\alpha)$ and the associated baseline is

$p_0 = \psi_i(\alpha)/(1-\alpha)$. *If* $A_{ij} \neq 0$ *for at least one pair,* $\mathcal{A}$ *contains at most one element.*

Proof. Necessity follows by applying Theorem 6.2 to every pair and observing that the model gives $\psi_i(\alpha) = (1-\alpha)p_0$ for every i.

For sufficiency, fix $\alpha \in \mathcal{A}$. The restriction $\Delta_{ij} = \alpha A_{ij}$ rearranges to

$$D_j(\alpha)P_j - D_i(\alpha)P_i = \alpha(q_j - q_i)|_S,$$

so $\psi_i(\alpha) = \psi_j(\alpha)$ for every pair. Let $\psi(\alpha)$ be their common value and set $p_0 = \psi(\alpha)/(1-\alpha)$. Summing any $\psi_i(\alpha)$ over S gives $1-\alpha$, and nonnegativity therefore places p_0 in $\Delta(S)$. By construction, every regime satisfies the unnormalized model equation.

Finally, if one A_{ij} is nonzero and both α and α' belong to $\mathcal{A}$, then $(\alpha-\alpha')A_{ij} = 0$, so $\alpha = \alpha'$. □

C.2 Generic Identification

Identification is robust across exposure designs in the topological sense of McManus (1992). Fix a proper menu $S \subsetneq \mathcal{X}$ with $|S| = m \geq 2$ and let $q = (q_1, \ldots, q_K) \in \Delta(\mathcal{X})^K$ be an exposure design. For each parameter pair (α, p_0), let $\Phi_q(\alpha, p_0)$ collect the choice distributions generated across these regimes:

$$\Phi_q : (0,1) \times \Delta^\circ(S) \longrightarrow (\Delta(S))^K, \qquad \Phi_q(\alpha, p_0) = (P_1, \ldots, P_K).$$

Proposition C.3 (Generic identification with $K \geq 3$)**.** *For* $K \geq 3$*, there is an open dense set* $Q^* \subset \Delta(\mathcal{X})^K$ *such that, for every* $q \in Q^*$*, the map* Φ_q *is injective and its Jacobian with respect to* (α, p_0) *has full column rank* m *at every* $(\alpha, p_0) \in (0,1) \times \Delta^\circ(S)$.

Proof. For each regime write $v_i = q_i|_S$ and $\mu_i = \mathbf{1}^\top v_i$. Let Q^* contain the exposure designs for which the vectors $\{v_i\}_{i=1}^K$ do not all lie on one affine line. Equivalently, some pair of differences $v_j - v_i$ and $v_k - v_i$ is linearly independent. A nonzero two-by-two minor witnesses this condition, so Q^* is open in the relative topology.

To see density, first perturb any exposure design into the relative interior of $\Delta(\mathcal{X})^K$. If the restricted vectors remain collinear and are not all equal, choose $v_j - v_i \neq 0$ and a third index k. Move an arbitrarily small amount of exposure in regime k from an outside alternative to a coordinate direction in S that is not parallel to $v_j - v_i$. The perturbed design leaves the affine line.

If all restricted vectors are equal, perturb two different regimes in two different coordinate directions. Since $m \geq 2$ and $K \geq 3$, the resulting differences are linearly independent. Thus Q^* is dense.

We next show that an exposure design in Q^* cannot generate the same choice distribution in every regime. If $P_i = P$ for all i, the unnormalized model equation gives

$$v_i - v_j = (\mu_i - \mu_j)P$$

for every pair, so all v_i lie on one affine line, a contradiction.

For the rank claim, the derivative of P_i in a baseline direction δ satisfying $\mathbf{1}^\top \delta = 0$ is

$$D_{p_0} P_i[\delta] = \frac{1-\alpha}{(1-\alpha) + \alpha\mu_i}\,\delta.$$

These directions contribute rank $m-1$ in the stacked output. Differentiating with respect to α gives

$$\frac{\partial P_i}{\partial \alpha} = \frac{v_i - p_0 + (1-\mu_i)P_i}{(1-\alpha) + \alpha\mu_i}.$$

If the α direction belonged to the span of the baseline directions, there would be one tangent vector δ for which

$$v_i - p_0 + (1-\mu_i)P_i = (1-\alpha)\delta$$

for every i. Pairwise subtraction would give $A_{ij} = 0$ for all i, j. Theorem 6.2 would then imply $P_i = P_j$ for all pairs, contradicting the preceding paragraph. The Jacobian therefore has full rank m.

Finally, suppose two parameter pairs generate the same $(P_1, \ldots, P_K)$. Because the choices do not all coincide, some pair has $P_i \neq P_j$. Theorem 6.2 identifies the same unique α from that pair under both parameterizations. The unnormalized equation then recovers the same p_0. Hence Φ_q is injective. □

C.3 Partially Observed Exposure

Suppose (α, p_0) is known and the analyst observes the choice distribution P, while the exposure distribution q is unobserved or partly recorded. The observed P determines the frequencies $q|_S$ on the menu once the available share q_S is known.

Proposition C.4 (The exposure set generated by one choice distribution)**.** *Fix a proper menu $S \subsetneq \mathcal{X}$,*

$\alpha \in (0,1)$, $\kappa = (1-\alpha)/\alpha$, *and* $p_0 \in \Delta^\circ(S)$. *If* P *is generated by some exposure distribution, then* $P \in \Delta^\circ(S)$ *and the set of exposure distributions generating* P *is*

$$\left\{\tilde{q} \in \Delta(\mathcal{X}) : \tilde{q}|_S = (\kappa+\mu)P - \kappa p_0 \text{ for some } \mu \in [\underline{\mu}(P), 1]\right\},$$

where

$$\underline{\mu}(P) = \max\left\{0, \max_{x\in S} \kappa\left(\frac{p_0(x)}{P(x)} - 1\right)\right\}.$$

For each admissible μ*, the remaining mass* $1-\mu$ *may be allocated arbitrarily over* $\mathcal{X} \setminus S$.

Proof. Interiority follows from (1): $P(x) \geq (1-\alpha)p_0(x) > 0$. If $\tilde{q}_S = \mu$, equation (3) yields

$$\tilde{q}|_S = (\kappa+\mu)P - \kappa p_0.$$

The right side sums to μ. Its x coordinate is nonnegative precisely when

$$\mu \geq \kappa\left(\frac{p_0(x)}{P(x)} - 1\right).$$

Taking the maximum over coordinates and intersecting with $[0,1]$ gives the stated interval. For a fixed available exposure vector, the represented rule is independent of how the residual mass is allocated outside S, so every such outside allocation is admissible. □

For each admissible $\mu > 0$, the associated conditional exposure distribution is

$$\tilde{q}(\cdot \mid S) = P + \frac{\kappa}{\mu}(P - p_0),$$

so changing μ moves along the ray from the baseline through observed choice.

Corollary C.5 (Choice restrictions with calibrated parameters)**.** *For fixed* (α, p_0) *with a full-support baseline, a distribution* $P \in \Delta(S)$ *is generated by some exposure if and only if*

$$P(x) \geq (1-\alpha)p_0(x) \qquad \text{for every } x \in S.$$

Proof. Necessity follows from (1). For sufficiency set

$$q|_S = \frac{P - (1-\alpha)p_0}{\alpha}.$$

The vector is nonnegative by hypothesis and sums to one, so it defines an exposure distribution supported on S. Substitution returns P. □

Proposition C.6 (One exposure statistic restores point identification)**.** *Suppose $z = a^\top q|_S$ is observed for a known $a \in \mathbb{R}^S$ with $a^\top P \neq 0$. Then*

$$q_S = \frac{z - \kappa a^\top (P - p_0)}{a^\top P}, \qquad q|_S = (\kappa + q_S)P - \kappa p_0.$$

Proof. Apply $a^\top$ to the identity $q|_S = \kappa(P - p_0) + q_S P$ and solve for q_S. □

Direct observation of the available share corresponds to $a = \mathbf{1}$. Observing $q(x)$ for one available alternative x corresponds to taking $a = \delta_x$ and determines all frequencies $q|_S$ on the menu.

Corollary C.7 (Partially observed exposure)**.** *Suppose the analyst observes $\{q(x) : x \in B\}$ for some $B \subseteq \mathcal{X}$.*

(i) *If $B \cap S \neq \varnothing$, then $q|_S$ is point identified.*

(ii) *If $B \subsetneq \mathcal{X} \setminus S$, then the identified interval for q_S is $[\underline{\mu}(P), 1 - q(B)]$.*

(iii) *If $B = \mathcal{X} \setminus S$, then $q_S = 1 - q(B)$ and $q|_S$ is point identified.*

Proof. Part (i) applies Proposition C.6 to any observed coordinate in S. For parts (ii) and (iii), the unavailable share must satisfy $1 - q_S \geq q(B)$. An unobserved unavailable alternative permits every value up to $1 - q(B)$ in part (ii), while observing the whole complement forces equality in part (iii). □

Example C.8 (A binary identified set)**.** Let $S = \{x, y\}$, $p_0 = (0.4, 0.6)$, $\alpha = 1/2$, and $P = (0.6, 0.4)$. Then $\kappa = 1$, $\underline{\mu}(P) = 1/2$, and

$$q|_S(\mu) = (0.2 + 0.6\mu,\ 0.4\mu - 0.2), \qquad \mu \in [1/2, 1].$$

At $\mu = 1/2$, the frequency vector $q|_S$ is $(0.5, 0)$; at $\mu = 1$, it is $(0.8, 0.2)$. Both generate the same choice distribution. Observing either $q(x)$ or $q(y)$ selects a unique point.

D Heterogeneous Influence Parameters

Suppose consumers follow AWI on a common menu, receive the same exposure distribution in each treatment, and have stable population shares across treatments. Their aggregate choices sat-

isfy Exposure Reallocation Independence. When influence parameters differ, aggregate choices may violate Relative Choice Independence. Variation in the exposure distribution can reveal the distribution of influence parameters.

Fix a proper menu S. A consumer type is a pair $(\kappa, p_0) \in (0, \infty) \times \Delta(S)$, where $\kappa = (1-\alpha)/\alpha$ reparameterizes the influence parameter α as in (3). Larger κ means a smaller influence parameter. Types are distributed according to F, and F_κ denotes its marginal on the first coordinate. Assume

$$\operatorname{supp} F_\kappa \subseteq [\underline{\kappa}, \overline{\kappa}], \qquad 0 < \underline{\kappa} \leq \overline{\kappa} < \infty. \tag{31}$$

Every consumer in a treatment cell receives the same exposure, described by an available share $\mu \in (0, 1]$ and a conditional exposure distribution $r \in \Delta(S)$. Write $P_{\kappa,p_0}(\mu, r)$ for this type's choice distribution $p(\cdot \mid S; q)$ when $q_S = \mu$ and $q(\cdot \mid S) = r$. Equation (3) gives

$$P_{\kappa,p_0}(\mu, r) = \frac{\kappa p_0 + \mu r}{\kappa + \mu}, \tag{32}$$

and the analyst observes the aggregate $\overline{P}(\mu, r) = \int P_{\kappa,p_0}(\mu, r)\, dF(\kappa, p_0)$.

D.1 Identification from Pooled Data

Proposition D.1 (Aggregate response and the distribution of influence parameters)**.** *Let F satisfy* (31) *and define the average weight on the conditional exposure distribution*

$$\bar{b}(\mu) = \int \frac{\mu}{\kappa + \mu}\, dF_\kappa(\kappa), \qquad \mu \in (0, 1]. \tag{33}$$

(i) *For every $\mu \in (0, 1]$ and all $r, r' \in \Delta(S)$,*

$$\overline{P}(\mu, r) - \overline{P}(\mu, r') = \bar{b}(\mu)\,(r - r'), \tag{34}$$

and $\bar{b}' > 0$, $\bar{b}'' < 0$ on $(0, 1]$.

(ii) *Fix two distinct conditional exposure distributions $r^0, r^1 \in \Delta(S)$, and suppose $\overline{P}(\mu, r^0)$ and $\overline{P}(\mu, r^1)$ are observed for every μ in an open interval $I \subseteq (0, 1)$. Then F_κ is uniquely deter-*

mined. These observations identify

$$H(\mu) \equiv \frac{\bar{b}(\mu)}{\mu} = \int \frac{1}{\kappa + \mu}\, dF_\kappa(\kappa), \tag{35}$$

the Stieltjes transform of F_κ.

(iii) *The same observations determine* $m_x(\kappa) \equiv \mathbb{E}[p_0(x) \mid \kappa]$ *for every* $x \in S$ *and* F_κ*-almost every* κ.

Changing the conditional exposure distribution at a fixed available share leaves each type's baseline contribution unchanged, so it cancels when choices are differenced, even when the baseline and influence parameter are correlated. At a given available share, each type's weight on the conditional exposure distribution depends on its influence parameter. The average weight, $\bar{b}(\mu)$, is identified by two distinct conditional exposure distributions. Part (ii) identifies the distribution of κ from this average response over an open interval of available shares under the stated support conditions. Part (iii) then recovers the mean baseline distribution conditional on κ from the remaining choice levels.

D.2 Aggregation

Proposition D.2 (Aggregation)**.** *Let* F *satisfy* (31)*. Aggregate choice admits the representation*

$$\overline{P}(\mu, r) = \frac{\kappa_0 \bar{p}_0 + \mu r}{\kappa_0 + \mu} \tag{36}$$

for some $\kappa_0 > 0$ *and* $\bar{p}_0 \in \Delta(S)$*, at every* μ *in an open subinterval of* $(0, 1)$ *and every* r *in a set containing two distinct conditional exposure distributions, if and only if* F_κ *is degenerate at* κ_0*. When the representation holds,* $\bar{p}_0 = \mathbb{E}[p_0]$.

With a common influence parameter, aggregate choices follow AWI with the mean baseline distribution. Heterogeneity in the influence parameter changes how the average weight $\bar{b}(\mu)$ varies with the available share, so pooled data no longer satisfy the fixed-menu representation of Theorem 5.1.

D.3 Testing Aggregate Responses

Section 5 characterizes the exposure–choice rule of a homogeneous observational unit. With heterogeneous influence parameters, pooled choice need not satisfy Relative Choice Independence even though each individual does. Failure of that condition in aggregate data can therefore reflect either a different individual rule or heterogeneity in the influence parameter.

Under heterogeneous persistent redrawing, divide the average weight $\bar{b}(\mu)$ on the conditional exposure distribution by the available share. The resulting response per unit of raw exposure frequency is

$$H(\mu) = \frac{\bar{b}(\mu)}{\mu} = \int \frac{1}{\kappa + \mu}\, dF_\kappa(\kappa).$$

Thus H is the Stieltjes transform of a probability distribution. For every integer $m \geq 1$,

$$(-1)^m H^{(m)}(\mu) = m! \int \frac{1}{(\kappa + \mu)^{m+1}}\, dF_\kappa(\kappa) > 0. \tag{37}$$

In particular, H is strictly decreasing. One-shot fallback has $H_1(\mu) = \alpha$, so it cannot be exactly observationally equivalent to a mixture of persistent redrawers at two distinct available shares. The gap is quantitative. Jensen's inequality gives $-H'(\mu) \geq H(\mu)^2$, and hence

$$\frac{1}{H(\mu_2)} - \frac{1}{H(\mu_1)} \geq \mu_2 - \mu_1 \qquad (\mu_2 > \mu_1). \tag{38}$$

If $0 < \varepsilon < \alpha$ and the aggregate response H under persistent redrawing stays within ε of the constant α on an interval of length L, equation (38) requires

$$\varepsilon \geq \frac{\sqrt{1 + \alpha^2 L^2} - 1}{L}.$$

A positive one-shot response therefore cannot be approximated arbitrarily closely over a fixed nondegenerate share interval. More generally, a homogeneous rule with at most N attempts has

$$H_N(\mu) = \frac{b_N(\mu)}{\mu} = \alpha \sum_{t=0}^{N-1} [\alpha(1 - \mu)]^t,$$

a polynomial of degree $N - 1$. If H_N coincided with the aggregate response H under persistent redrawing on an open interval, their Nth derivatives would coincide there. The derivative of H_N is zero at that order, whereas equation (37) is strictly signed. No finite retry cap, including one-shot

fallback, therefore belongs exactly to the mixture class on an open interval.

On a finite set of exposure treatments, a large retry cap can produce choice probabilities close to those under persistent redrawing. Let b_∞ denote the weight on the conditional exposure distribution under homogeneous persistent redrawing. Equation (30) gives the uniform bound

$$0 \le b_\infty(\mu) - b_N(\mu) = b_\infty(\mu)[\alpha(1-\mu)]^N \le \alpha^N, \qquad \mu \in [0,1].$$

Large retry caps can therefore be close to persistent redrawing throughout the observable share range. With finitely many treatments, violations of the restrictions above can reject the mixture model. The ability to distinguish a large retry cap from persistent redrawing depends on the chosen available shares and sampling precision.

If H is generated by a persistent-redrawing mixture satisfying (31), observations on an open interval identify the unique distribution F_κ. Other mechanisms generating the same H are observationally equivalent with respect to these aggregate responses. If no admissible probability distribution generates the observed H, the persistent-redrawing mixture model is rejected.

D.4 Proofs

Proof of Proposition D.1. Integrating (32) over types gives

$$\overline{P}(\mu, r) = \int \frac{\kappa p_0}{\kappa + \mu}\, dF(\kappa, p_0) + \bar{b}(\mu)\, r, \tag{39}$$

where the first term does not depend on r. Subtracting the same expression at r' proves (34). On $[\underline{\kappa}, \overline{\kappa}]$ with $\underline{\kappa} > 0$, the integrand of (33) and its first two derivatives in μ are bounded uniformly, so differentiation passes under the integral:

$$\bar{b}'(\mu) = \int \frac{\kappa}{(\kappa+\mu)^2}\, dF_\kappa > 0, \qquad \bar{b}''(\mu) = -\int \frac{2\kappa}{(\kappa+\mu)^3}\, dF_\kappa < 0.$$

This proves part (i).

For part (ii), choose $x \in S$ with $r^1(x) \neq r^0(x)$, which exists because the two conditional exposure distributions differ. Equation (34) gives

$$\bar{b}(\mu) = \frac{\overline{P}_x(\mu, r^1) - \overline{P}_x(\mu, r^0)}{r^1(x) - r^0(x)},$$

so $\bar{b}$, and hence $H = \bar{b}/\mu$, is observed on I. The support condition permits differentiation under the integral at every order and gives (37). Because F_κ is supported in $[\underline{\kappa}, \overline{\kappa}]$, the map $z \mapsto \int (\kappa + z)^{-1} dF_\kappa(\kappa)$ is analytic on the complement of $[-\overline{\kappa}, -\underline{\kappa}]$, and in particular on a complex neighborhood of $(-\underline{\kappa}, \infty)$. Its values on the open interval I therefore determine it everywhere on that domain by analytic continuation. A Stieltjes transform determines its measure, so F_κ is unique.

For part (iii), subtract $\bar{b}(\mu)r$ from (39) and read coordinate x:

$$R_x(\mu) \equiv \overline{P}_x(\mu, r) - \bar{b}(\mu) r(x) = \int \frac{\kappa\, m_x(\kappa)}{\kappa + \mu}\, dF_\kappa(\kappa).$$

Since m_x takes values in $[0, 1]$ and $\kappa \leq \overline{\kappa}$, the measure $\kappa\, m_x(\kappa)\, dF_\kappa(\kappa)$ is finite and positive, and R_x is its Stieltjes transform. The argument of part (ii) determines that measure from its values on I. Part (ii) already determines F_κ, and $\kappa \geq \underline{\kappa} > 0$, so m_x is determined F_κ-almost everywhere. □

Proof of Proposition D.2. Suppose F_κ is degenerate at κ_0. Then $\bar{b}(\mu) = \mu/(\kappa_0 + \mu)$, and the first term of (39) equals $\kappa_0 \mathbb{E}[p_0]/(\kappa_0 + \mu)$. Substituting both into (39) gives (36) with $\bar{p}_0 = \mathbb{E}[p_0]$.

Conversely, suppose (36) holds on an open subinterval I and at two distinct conditional exposure distributions. Applying part (i) of Proposition D.1 to the left side and differencing the right side at those two conditional exposure distributions gives $\bar{b}(\mu) = \mu/(\kappa_0 + \mu)$ for every $\mu \in I$, so $H(\mu) = (\kappa_0 + \mu)^{-1}$ there. The uniqueness established in part (ii) identifies F_κ with the point mass at κ_0. Matching the terms in (39) that do not involve r then gives $\bar{p}_0 = \mathbb{E}[p_0]$. □

E Sequential Exposure

Suppose the consumer applies AWI to the empirical distribution of the influencer's choices observed so far.

Fix S, $p_0 \in \Delta(S)$, and $\alpha \in (0, 1)$, and let $\kappa = (1-\alpha)/\alpha$. The consumer observes the influencer's choices $Z_1, Z_2, \ldots \in \mathcal{X}$. After t observations, define

$$N_t(x) = \sum_{n=1}^{t} \mathbf{1}\{Z_n = x\}, \qquad N_{t,S} = \sum_{x \in S} N_t(x), \qquad \widehat{q}_t = \frac{1}{t} \sum_{n=1}^{t} \delta_{Z_n}.$$

Write $P_t = p(\cdot \mid S; \widehat{q}_t)$ for the consumer's choice distribution after these t observations. Applying equation (3) to $\widehat{q}_t$ gives

$$P_t = \frac{\kappa p_0 + \widehat{q}_t|_S}{\kappa + \widehat{q}_{t,S}} = \frac{t\kappa p_0 + N_t|_S}{t\kappa + N_{t,S}}. \tag{40}$$

The menu, baseline distribution, and influence parameter remain fixed as the empirical exposure distribution $\widehat{q}_t$ changes.

Proposition E.1 (Updating after one more observation)**.** *Let* $w_t = t\kappa + N_{t,S}$.

(i) P_t *depends on the history only through its category counts.*

(ii) *If* $Z_{t+1} \notin S$*, then*

$$P_{t+1} = \frac{w_t P_t + \kappa p_0}{w_t + \kappa}.$$

Whenever $P_t \neq p_0$*, the new observation moves choice strictly toward the baseline.*

(iii) *If* $Z_{t+1} = x \in S$*, then*

$$P_{t+1} = \frac{w_t P_t + \kappa p_0 + \delta_x}{w_t + \kappa + 1}.$$

(iv) *If the observations are independent draws from a fixed distribution* q*, then* $P_t \to p(\cdot \mid S; q)$ *almost surely.*

Proof. Part (i) follows from the count representation in (40). Since $w_t P_t = t\kappa p_0 + N_t|_S$, the numerator at date $t+1$ is

$$w_t P_t + \kappa p_0 + \left(N_{t+1}|_S - N_t|_S\right).$$

The final increment equals zero when $Z_{t+1} \notin S$ and equals δ_x when $Z_{t+1} = x \in S$. The denominator rises by κ in the first case and by $\kappa + 1$ in the second, which proves parts (ii) and (iii). In part (ii), the weight on p_0 is $\kappa/(w_t + \kappa) \in (0, 1)$, so the movement is strict whenever $P_t \neq p_0$. For part (iv), the strong law gives $\widehat{q}_t \to q$ coordinatewise, and the map in equation (3) is continuous because its denominator is bounded below by $\kappa > 0$. □

Observing an unavailable alternative leaves the counts of available alternatives unchanged and reduces their share of all observations whenever that share is positive. The consumer's choice distribution therefore moves toward her baseline. Observing an available alternative increases its count by one. The coefficient on the baseline also increases by κ, because the empirical exposure distribution uses the updated number of observations. The weights on the added baseline term and, for an available observation, the observed alternative shrink as w_t grows.

These updates use the observed sequence of influencer choices. Appendix F.5 considers how to locate changes in the distribution generating that sequence.

F Estimation

A minimum-distance estimator uses samples of exposure and consumer choices to estimate the common baseline distribution and influence parameter. The moment conditions preserve the unnormalized structure of equation (14), so the baseline enters linearly once the influence parameter is fixed.

F.1 Moment Conditions

Fix a menu S with $m = |S| \geq 2$ and observe K exposure regimes. In regime i, let q_i be the exposure distribution, $v_i = q_i|_S$, $\mu_i = q_{i,S}$, and P_i the population choice distribution. A sample of N_i independent choices produces empirical frequencies $\widehat{P}_i$. Define $D_i(\alpha) = (1 - \alpha) + \alpha\mu_i$. Equation (14) suggests the sample moment

$$m_i(\alpha, p_0) = D_i(\alpha)\widehat{P}_i - (1 - \alpha)p_0 - \alpha v_i. \tag{41}$$

Let α_0 denote the true influence parameter. At the true parameters, $\mathbb{E}[m_i(\alpha_0, p_0)] = 0$ for every regime.

The coordinates of each moment sum to zero. Index the alternatives in S by $1, \ldots, m$, and let

$$H = \begin{pmatrix} I_{m-1} & 0_{(m-1)\times 1} \end{pmatrix}$$

drop the last coordinate. Write $\widetilde{m}_i = Hm_i$, $\widetilde{p}_0 = Hp_0$, and stack the reduced moments into

$$\widetilde{m}(\theta) = \left(\widetilde{m}_1(\theta)^\mathsf{T}, \ldots, \widetilde{m}_K(\theta)^\mathsf{T}\right)^\mathsf{T}, \qquad \theta = (\alpha, \widetilde{p}_0).$$

There are $K(m - 1)$ moment conditions and m free parameters.

F.2 Concentrating Out the Baseline

For a positive-definite block-diagonal weight $W = \mathrm{blkdiag}(W_1, \ldots, W_K)$, consider

$$Q_N(\theta; W) = N\,\widetilde{m}(\theta)^\mathsf{T} W \widetilde{m}(\theta), \qquad N = \sum_{i=1}^{K} N_i.$$

At a fixed α, the criterion is quadratic in $\widetilde{p}_0$. Its minimizer is

$$\widehat{\widetilde{p}}_0(\alpha) = \frac{1}{1-\alpha}\left(\sum_{i=1}^{K} W_i\right)^{-1} \sum_{i=1}^{K} W_i H\big[D_i(\alpha)\widehat{P}_i - \alpha v_i\big]. \tag{42}$$

Estimation can therefore proceed through the one-dimensional concentrated criterion

$$Q_N^c(\alpha; W) = Q_N\big((\alpha, \widehat{\widetilde{p}}_0(\alpha)); W\big).$$

Fix $0 < \varepsilon < 1/2$ such that $\alpha_0 \in (\varepsilon, 1-\varepsilon)$. After minimizing over $[\varepsilon, 1-\varepsilon]$ for α, recover the last coordinate of p_0 by making its probabilities sum to one. To impose the simplex constraint in finite samples, minimize the same criterion over $(\alpha, p_0) \in [\varepsilon, 1-\varepsilon] \times \Delta(S)$.

F.3 Sampling Variation and Weighting

Let $\pi_i = N_i/N$. Under multinomial sampling,

$$\mathrm{Var}(\widehat{P}_i) = \frac{1}{N_i}\Sigma(P_i), \qquad \Sigma(P_i) = \mathrm{diag}(P_i) - P_i P_i^{\mathsf{T}}.$$

Hence

$$\Omega = \mathrm{Var}\big(\sqrt{N}\,\widetilde{m}(\theta_0)\big) = \mathrm{blkdiag}\left(\frac{D_1(\alpha_0)^2}{\pi_1} H\Sigma(P_1)H^{\mathsf{T}}, \ldots, \frac{D_K(\alpha_0)^2}{\pi_K} H\Sigma(P_K)H^{\mathsf{T}}\right).$$

A two-step estimator first minimizes with a convenient preliminary weight, constructs a plug-in estimate $\widehat{\Omega}$, and then minimizes with $W = \widehat{\Omega}^{-1}$.

Let $g(\theta) = \mathbb{E}[\widetilde{m}(\theta)]$ and let G be its Jacobian at the true parameter. The block derivatives for regime i are

$$\frac{\partial \mathbb{E}[\widetilde{m}_i]}{\partial \widetilde{p}_0^{\mathsf{T}}} = -(1-\alpha) I_{m-1}, \qquad \frac{\partial \mathbb{E}[\widetilde{m}_i]}{\partial \alpha} = H\big[(\mu_i - 1)P_i + p_0 - v_i\big].$$

Under the usual regularity and rank conditions,

$$\sqrt{N}(\widehat{\theta} - \theta_0) \Rightarrow \mathcal{N}\left(0, (G^{\mathsf{T}}\Omega^{-1}G)^{-1}\right).$$

The rank condition is the sample analogue of informative adjusted exposure variation in Section 6.

F.4 Estimated Exposure

Suppose regime i also contains T_i independent observations drawn from q_i, producing the empirical exposure distribution $\widehat{q}_i$. Let $\widehat{v}_i = \widehat{q}_i|_S$, $\widehat{\mu}_i = \mathbf{1}^\top \widehat{v}_i$, and $\widehat{D}_i(\alpha) = (1-\alpha) + \alpha\widehat{\mu}_i$. Replace (41) by

$$\widehat{m}_i(\alpha, p_0) = \widehat{D}_i(\alpha)\widehat{P}_i - (1-\alpha)p_0 - \alpha\widehat{v}_i.$$

Independence of the exposure and choice samples gives the same population moment as before. The extra sampling variation changes the covariance matrix.

Write $\delta P_i = \widehat{P}_i - P_i$, $\delta v_i = \widehat{v}_i - v_i$, and $J_i = P_i\mathbf{1}^\top - I_m$. At the true parameter,

$$\widehat{m}_i(\theta_0) = D_i(\alpha_0)\delta P_i + \alpha_0 J_i \delta v_i + \alpha_0(\mathbf{1}^\top \delta v_i)\delta P_i.$$

The final term is of smaller order. Since

$$\mathrm{Var}(\delta v_i) = \frac{1}{T_i}\Sigma_i^q, \qquad \Sigma_i^q = \mathrm{diag}(v_i) - v_i v_i^\top,$$

the leading covariance of the unreduced moment is

$$\mathrm{Var}\big(\widehat{m}_i(\theta_0)\big) = \frac{D_i(\alpha_0)^2}{N_i}\Sigma(P_i) + \frac{\alpha_0^2}{T_i} J_i \Sigma_i^q J_i^\top + O\big((N_i T_i)^{-1}\big).$$

Premultiplying and postmultiplying by H gives the block used in the optimal weight. When T_i/N_i grows without bound, the known-exposure covariance is recovered.

F.5 Locating Regimes in a Stream

The moments above take the partition of the data into regimes as given. The influencer's choices are observed as a sequence, and the boundaries then have to be located before the structural moments can be applied. Suppose observations are independent categorical draws with one break,

$$Z_t \sim q^{(1)} \ \text{ for } t \le \tau, \qquad Z_t \sim q^{(2)} \ \text{ for } t > \tau,$$

where $q^{(1)} \neq q^{(2)}$, the break date τ stays away from the endpoints, and both segment lengths diverge. A first stage estimates $(\widehat{\tau}, \widehat{q}^{(1)}, \widehat{q}^{(2)})$ by segmentation. Penalized-likelihood procedures locate an unknown number of breaks at linear computational cost (Killick et al., 2012), and change-

point tests built for high-dimensional multinomial data apply directly to observation streams (Wang et al., 2018). The second stage feeds the estimated segments into the moment conditions above.

Break-dating error is negligible at first order. Suppose the estimator satisfies $\widehat{\tau} - \tau = O_p(1)$, as fixed-magnitude break estimators do. Replacing the true split by $\widehat{\tau}$ then misclassifies $O_p(1)$ observations, so the resulting error in each segment frequency is $O_p(T_i^{-1}) = o_p(T_i^{-1/2})$. The estimated break date therefore leaves the first-order limit of the empirical exposure frequencies unchanged.

Applying the identification result across the two segments requires a stable menu, baseline distribution, and influence parameter, as in Assumption 6.1. Choice observations must correspond to the exposure received during each segment.

F.6 Specification Tests and Weak Identification

The model is overidentified when $K(m-1) > m$. With the optimal two-step weight,

$$J_N = Q_N(\widehat{\theta}; \widehat{\Omega}^{-1}) \Rightarrow \chi^2_{K(m-1)-m}$$

under correct specification. The test asks whether all regimes can be reconciled with one baseline and one influence parameter.

When the adjusted exposure movements are small, the influence parameter may be weakly identified. At each fixed α, the criterion remains quadratic in the baseline probabilities. The concentrated criterion provides an identification-robust confidence set. Fix a significance level $\gamma \in (0, 1)$. For each candidate α, estimate the baseline by (42) using a covariance matrix valid under that candidate and retain the values satisfying

$$Q_N^c(\alpha; \widehat{\Omega}(\alpha)^{-1}) \leq \chi^2_{(K-1)(m-1),1-\gamma}.$$

At the true α, the minimized criterion tests the equality of the K regime-specific baseline moments and has $(K-1)(m-1)$ degrees of freedom. Inverting that test yields asymptotically valid coverage as the adjusted exposure movement approaches zero (Stock and Wright, 2000).

G A Relative-Entropy Representation

For $q_S > 0$, equation (3) minimizes a weighted sum of relative entropies. For distributions $r, \pi \in \Delta(S)$, write $D_{\mathrm{KL}}(r\|\pi) = \sum_{x\in S} r(x)\log[r(x)/\pi(x)]$. A term is zero when $r(x) = 0$ and is $+\infty$ when $r(x) > 0$ and $\pi(x) = 0$. Relative entropy is also used in choice models by Maccheroni et al. (2006). Here,

$$p(\cdot \mid S; q) = \arg\min_{\pi\in\Delta(S)} \left\{\kappa D_{\mathrm{KL}}\big(p_0(\cdot \mid S)\|\pi\big) + q_S D_{\mathrm{KL}}\big(q(\cdot \mid S)\|\pi\big)\right\}.$$

Expanding the objective, the Lagrange conditions give $\pi(x) = [\kappa p_0(x \mid S) + q(x)]/(\kappa + q_S)$ for every $x \in S$, which is equation (3). Strict convexity on the relevant support gives uniqueness.

The two terms compare a candidate choice distribution with the baseline distribution and the conditional exposure distribution. Their coefficients are κ and q_S. Holding the conditional exposure distribution fixed, a larger available share places more weight on that distribution in the objective.